Journal of Energy Storage 154 (2026) 121189

Contents lists available at ScienceDirect

Journal of Energy Storage

journal homepage: www.elsevier.com/locate/est

Research papers

Experimental evaluation of copper slags as a thermal energy storage medium in axial and radial packed-bed devices

Felipe G. Battisti [a],*, Ignacio Calderón-Vásquez [b], Jose Ortega [b], Ian Wolde [b], Valentina Segovia [b], Raimundo Claren [b], Ignacio Arias [b], Nicolás Pailahueque [c], Rodrigo A. Escobar [b,d], José M. Cardemil [b,d]

[a] Departamento de Mecánica, Facultad de Ingeniería, Universidad Tecnológica Metropolitana, José Pedro Alessandri 1242, Santiago, Chile
[b] Departamento de Ingeniería Mecánica y Metalúrgica, Escuela de Ingeniería, Pontificia Universidad Católica de Chile, Vicuña Mackenna 4860, Santiago, Chile
[c] Departamento de Ingeniería Química y Bioprocesos, Universidad de Santiago de Chile, Libertador Bernardo OHiggins 3363, Santiago, Chile
[d] Centro de Energía UC, Centro del Desierto de Atacama, Pontificia Universidad Católica de Chile, Santiago, Chile

ARTICLE INFO

Dataset link: https://doi.org/10.17632/fgn79fzg7x.1

Keywords:
Thermal energy storage
Packed bed
Copper slag
Experimental assessment

ABSTRACT

Packed-bed thermal energy storage (PBTES) using industrial by-products, such as copper slag, provides a cost-effective and high-temperature solution for integrating variable renewables and decarbonizing industrial heat. This study presents a novel experimental comparison between axial (A-PBTES) and radial (R-PBTES) PBTES configurations with copper slag as the storage medium and air as the heat transfer fluid. Duplicate tests considered 1.5 h and 3.0 h charging intervals and airflow temperatures of 150 °C and 300 °C. Because the axial and radial units correspond to different bed volumes, geometries, and porosities, the reported results should be interpreted as a prototype-level comparison under common boundary conditions rather than a one-to-one topology ranking. The temperature profiles obtained from the A-PBTES revealed a clear axial thermocline. The average energy storage capacity achieved was 2.2 ± 0.7 kWh, and the discharge energy delivered to the airflow averaged 2.1 ± 0.7 kWh, resulting in a round-trip thermal efficiency of 93.5 ± 1.5%. The R-PBTES exhibited peak temperatures in the central region, with azimuthal asymmetry decreasing over time. It averaged an energy storage capacity of 0.7 ± 0.2 kWh and delivered 0.5 ± 0.2 kWh during discharge, resulting in a thermal efficiency of 73.9 ± 0.7%. It also showed faster energy exchange and reduced pressure drop. Finally, the thermal power profiles showed that the R-PBTES delivers and stores thermal energy faster than the A-PBTES. The latter configuration also maintains superior thermal stratification, achieving up to 14.5 % higher specific stored exergy compared to the R-PBTES, which experiences higher thermal dispersion. These findings confirm copper slag as a technically viable and low-cost TES material and highlight the radial topology's potential for reduced pressure drops and self-insulation, supporting slag-based PBTES for cost-effective, scalable applications in high-temperature energy storage, industrial process heat, and waste-heat recovery.

1. Introduction

Renewable electricity generation has increased substantially worldwide, reaching 26.5 % of total generation when hydropower is included [1]. This growth reflects sustained policy commitments by governments and international agencies to reshape the electricity portfolio in response to climate objectives [2]. Nevertheless, broader deployment of renewable resources remains constrained by the limited availability of reliable and cost-effective energy storage systems [3]. Because variable renewable energy (VRE) exhibits intermittent output, storage is essential for ensuring adequacy, flexibility, and grid reliability: photovoltaic (PV) plants are commonly coupled with battery energy storage systems (BESS) for short-duration balancing, whereas solar-thermal plants couple collectors with thermal energy storage (TES). In particular, TES is a key enabler of sector coupling (power–heat–chemical), facilitating greater utilization of renewable electricity and improving overall system efficiency.

Thermal energy storage (TES) mitigates the temporal mismatch between variable renewable generation and end-use demand across the residential, commercial, and industrial sectors [4]. By enabling sector coupling and integrating the power system with district heating/cooling and industrial processes, TES increases renewable utilization, enhances operational flexibility [5], and supports the decarbonization of high-temperature processes [6]. Within the electricity sector, TES can buffer heat produced by concentrating solar power (CSP) as well

* Corresponding author.
E-mail address: felipegbattisti@utem.cl (F.G. Battisti).

https://doi.org/10.1016/j.est.2026.121189
Received 24 October 2025; Received in revised form 9 January 2026; Accepted 16 February 2026
Available online 2 March 2026
2352-152X/

Abbreviations

A-PBTES	Axial packed-bed thermal energy storage
CFD	Computational fluid dynamics
CSPB	Copper slag packed bed
CSP	Concentrating solar power
FT	Flow transducer
HTF	Heat transfer fluid
nRMSE	Normalized root-mean-square error
PBTES	Packed-bed thermal energy storage
PT	Pressure transducer
PV	Photovoltaic
R-PBTES	Radial packed-bed thermal energy storage
RMSE	Root-mean-square error
TES	Thermal energy storage
TT	Temperature transducer

Nomenclature

Δp	Pressure drop [Pa]
ΔT	Temperature difference [K]
$\dot{Q}$	Thermal power (heat rate) [kW]
η	Efficiency [-]
ρ	Density [$\mathrm{kg\,m^{-3}}$]
τ	Normalized time [-]
ε	Bed porosity [-]
$V\!\!\!/$	Volume [$\mathrm{m^3}$]
c	Specific heat capacity [$\mathrm{kJ\,kg^{-1}\,K^{-1}}$]
D	Diameter [m]
ex	Specific exergy [$\mathrm{kJ\,kg^{-1}}$]
H	Height [m]
I_a	Asymmetry index [-]
k	Thermal conductivity [$\mathrm{W\,m^{-1}\,K^{-1}}$]
m	Mass [kg]
N	Number of elements, nodes [-]
Q	Energy stored [kW h]
r	Radius, radial coordinate [m]
T	Temperature [°C]
t	Time [s]
z	Axial coordinate [m]

Subscripts

0	Dead state
amb	Ambient
$bottom$	Lower end of the bed
cha	Charging stage
dis	Discharging stage
eff	Effective
i	i-th node, element
in	Inlet
$inner$	Inner radius
j	j-th circumferential node
out	Outlet
$outer$	Outer radius
$roundtrip$	Round trip
$slag$	Copper slag (solid medium)
$upper$	Upper end of the bed

as power-to-heat pathways (e.g., resistive heaters or heat pumps), thereby maintaining a dispatchable heat supply for power cycles and thermal networks [7,8]. Among commercially available alternatives, two-tank systems using molten salts are considered the state of the art for high-temperature processes. However, commonly used nitrate salt mixtures have an upper thermal stability near 600 °C and freezing points around 220 °C, restricting practical operation to approximately 250 °C–585 °C [9]. These limits constrain conversion efficiency and contribute to higher costs and operational complexity, including stringent control and safety measures to prevent solidification and leakage [10].

In this context, packed-bed thermal energy storage (PBTES) constitutes a promising alternative to conventional two-tank molten-salt systems. PBTES stores thermal energy commonly in a cylindrical container, where a bed of solid particles is disposed, which serves as the storage medium and comes into direct contact with the heat-transfer fluid (HTF) [11]. The packed-bed configuration promotes the formation of a thermocline – a stable temperature gradient – between the hot and cold regions of the bed [12].

Single-tank implementations of PBTES systems, particularly when paired with high-density filler materials, can reduce capital costs by approximately 33 % relative to two-tank molten-salt configurations [13,14]. When solid filler materials are employed, thermal energy is stored predominantly as sensible heat, $Q = mc\Delta T$, proportional to the filler mass m, specific heat capacity c, and the temperature swing ΔT between the initial and final states.

Consequently, the main advantages of packed-bed systems commonly include:

- low purchase cost of storage material,
- wide working temperature range,
- direct heat transfer between the HTF and the storage material (eliminating the need for intermediate heat exchangers),
- chemical stability with limited degradation and corrosion [11],
- possibility of using several alternative and low-cost HTFs, such as air [15]

Despite the advantages of PBTES, the main limitation of using solid sensible storage materials is their relatively low thermal capacity. This drawback can be mitigated by operating over large temperature swings and selecting high-density materials, which increases the volumetric energy density of the systems, particularly interesting for high-temperature applications [3].

Using solid storage media usually enables TES systems to operate at elevated temperatures, thereby increasing the availability of high-grade heat and improving exergy efficiency in downstream energy conversion processes. The design of TES units must observe operating conditions compatible with the target high-temperature applications; nevertheless, only a few commercially available technologies currently operate above 600 °C [6], using rocks [16], sand [17], and concrete [18].

Researchers have explored alternative vessel topologies to improve the thermal performance of packed-bed TES. The most common topology of PBTES is axial-flow thermocline tanks [19,20]. Zanganeh et al. proposed an inverted truncated-cone tank to mitigate creep-related shell deformations [11]. However, this geometry did not confer measurable thermal-performance gains relative to axial thermocline designs [21]. Leveraging the natural angle of repose of rock fill, Erasmus et al. examined a truncated-conical bed in which the smaller top diameter functions as the hot port and the larger bottom diameter forms the cold zone, aligning the bed's shape with the granular material's stable slope [22]. Beyond conical vessels, Trevisan et al. introduced a radial-flow topology in which the packed bed occupies the annulus between two concentric cylinders, forcing radial flow across the bed. This configuration produces a "self-insulating" temperature field that limits external losses and reduces pressure drop, but it exhibited lower thermal efficiency (≈72%) than conventional axial thermocline geometries (≈95%) [23].

The operation of an axial packed-bed thermal energy storage (A-PBTES) system comprises two stages: charging and discharging phases. During charging, the heat-transfer fluid (HTF) enters at one end of the vessel and advances along the axis, transferring heat to the granular filler by direct contact. Analogously, during the discharging phase, the HTF flows through the tank in the opposite direction, recovering the energy stored in the packed bed of rocks and leaving the tank at a higher temperature [24]. Depending on the integration with the thermal load, the HTF may circulate in an open loop (vented to the environment) or a closed loop (returned to the heat source/system).

In the packed bed, the HTF flows through the interstitial voids within the filler material, describing a tortuous flow. Convection between the fluid and the particles governs the heat exchange primarily with additional conduction across particle–to-particle contacts and through the tank wall. Although present, radiative transfer only becomes significant at operating temperatures above 800 °C [25].

In a radial packed-bed thermal energy storage (R-PBTES) system, the HTF flows from the bed's center towards the wall during the charging process, and in the reverse direction during the discharging phase. Compared to A-PBTES, the radial topology constitutes a self-insulating configuration resulting from the high-temperature zone in the packed bed's core, reducing the thermal losses to the environment, due to the smaller temperature gradient between the wall and surroundings [15,23]. Moreover, numerical studies have already shown that the radial topology decreases the overall pressure drop experienced by the HTF, e.g., Refs. [26,27].

1.1. Existing experimental installations

Given the potential benefits of packed-bed thermal energy storage (PBTES), the literature reports numerous experimental implementations. Fig. 1 compiles the reported studies across the range of scales and classifies them by bed diameter, height, flow topology, and HTF type. Most reported setups are on a laboratory scale, with relatively few on a pilot scale. Experiments typically employ a gaseous HTF with solid fillers; air is the most common choice because it is inexpensive, inert, and thermally and chemically stable [20,28].

Bed orientation and airflow direction are critical design variables for PBTES systems. Vertical configurations promote thermal stratification, a key parameter and determinant of storage efficiency [29], particularly observed in slender tanks with aspect ratio $H/D > 1$ (Height/Diameter). However, slender vessels exhibit higher internal stresses [13]; at large capacities, thermal expansion of the filler particles under cyclic operation can induce thermo-mechanical deformation and thermal ratcheting [23]. Accordingly, the studies summarized in Fig. 1 span a range of aspect ratios to elucidate the trade-off between thermal performance and structural response. A horizontal bed layout has been proposed to mitigate ratcheting-related stresses [30] while reducing cost and simplifying piping [31]. However, axial horizontal configurations tend to promote tunneling in the bed structure, impacting the thermal efficiency [32]. Finally, only a few experimental systems implement radial flow PBTES systems: lab-scale R-PBTES prototypes at KTH Royal Institute of Technology [23] and Sandia National Laboratories [33] with capacities of 50 kW h_{th} and 100 kW h_{th}, respectively.

1.2. Filler materials for PBTES systems

The literature reports a broad range of solids considered as filler materials for PBTES systems, including natural rocks [45], silica and magnesia firebricks [46], other ceramics [35], and industrial by-products [7,47–49]. These alternatives exhibit different advantages and limitations. While some of the materials mentioned above are straightforward to implement in PBTES systems, low-cost materials commonly exhibit low thermal storage capacity. On the other hand, materials that combine high storage capacity and thermochemical stability-such as advanced ceramics- tend to be more expensive. Compared to liquid media widely implemented in commercial applications (e.g., molten salts, thermal oils, water), many solid fillers that are usually studied exhibit lower thermal storage capacity.

Despite this trend, recent studies have shown the potential of industrial by-products derived from metallurgical processes, as shown in Table 1. These industrial residues frequently maintain thermal stability across a wide temperature range, enabling the possibility of operating at temperatures higher than 800 °C [50]. Slags from processes such as copper pyrometallurgy or iron and steel metallurgical processes have been considered good candidates as filler materials in thermal storage systems, due to their high thermal capacity and tolerance to high-temperature thermal cycling [47,50,51].

Several authors have reported the potential benefits of using industrial by-products as filler material in PBTES systems. Ortega-Fernández et al. [52] characterized the thermophysical and structural properties of electric arc furnace (EAF) slags and reported thermal stability up to 1100 °C. Agalit et al. [7] evaluated the induction furnace slags (IFS), finding that the specific heat capacity ranges from 0.463 $kJ\,kg^{-1}\,K^{-1}$ at 35 °C to 0.738 $kJ\,kg^{-1}\,K^{-1}$ for temperatures between 300 and 400 °C. In addition, the IFS slags remained thermally stable under thermal cycling between room temperature and 1000 °C. Lai et al. [36] examined the chemical composition and thermophysical properties of three typical coal slags, reporting c_p values that increase from 0.70 $kJ\,kg^{-1}\,K^{-1}$ at 10 °C to 1.0 $kJ\,kg^{-1}\,K^{-1}$ at 380 °C, with thermal stability up to 1100 °C. In particular, Ref. [25] analyzed the potential use of copper slag from Chilean foundries, considering its high availability and almost negligible cost. The material exhibited specific heat capacities comparable to molten salts, constituting a cost-efficient alternative for thermal storage.

Beyond material characterization at laboratory scale, recent studies have advanced modeling, operation, and application of PBTES devices using copper slags as the storage medium. Calderón-Vásquez et al. [57] developed a reduced-order parameter-fitted model for a horizontal air–solid PBTES calibrated against experiments that achieved a normalized root-mean-square error (RMSE) of about 4 % for estimating the state of charge. The analysis concluded that nearly 60 % of the discharged energy was recovered from the tank walls, despite not explicitly modeling them. The resulting combination of simplicity and accuracy makes such reduced-order formulations well-suited for integration into the control schemes for real systems.

Regarding monitoring the thermal state of cylindrical PBTES devices, Ref. [58] proposed an operational framework that relies exclusively on the top and the bottom bed temperatures. The resulting state metric provides a rigorous, non-arbitrary charge cut-off criterion and correlates required charging time and cut-off temperature with design conditions; thereby, it is useful for supervisory control and performance guarantees.

Furthermore, acknowledging the "as-received" variability of metallurgical residues, Ref. [59] developed a probabilistic model for the specific heat of Chilean copper slags, quantifying how c_p percentiles shift thermocline propagation during charge, and showing that round-trip efficiency is influenced more by the system's design features than by c_p dispersion. These results support using copper slag in PBTES when prioritizing design choices, such as bed geometry and flow management.

Regarding the application of PBTES systems, Ref. [60] reports a parametric study coupling an air flat-plate solar collector with a copper-slag PBTES for food drying. In that context, early flow-switching combined with smaller TES volumes and a single collector can reduce levelized cost by ∼40.5% versus a conventional configuration, highlighting the importance of stratification management and proper sizing of the TES system. In industrial contexts, Ref. [61] details the design and assessment of a concentrating solar thermal system that integrates a slag-filled packed bed, reinforcing the techno-economic feasibility of this waste-to-value pathway.

In summary, the use of slags from pyrometallurgical processes as storage material in TES systems is a promising alternative for the replacement of conventional materials, such as solar salts, due to the following advantages:

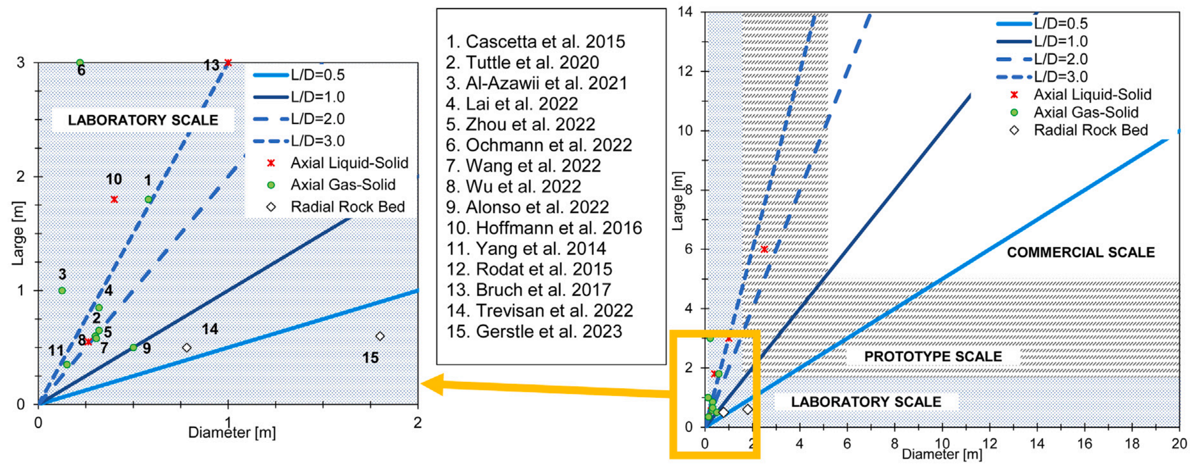


Fig. 1. Experimental PBTES systems reported in the literature, classified by dimension and scale [20,23,28,33–44]. *Source:* Modified from Esence et al. [12].

Table 1
Properties of different by-products used as thermal energy storage materials.

	Density [kg/m^3]	Specific heat capacity [kJ/kg K]	Thermal conductivity [W/mK]	Thermal stability [° C]	Cost [USD/t]	Ref.
EAF 1	3430	0.710–0.950	1.43	≤1100	80	[52]
EAF 2	3770	0.690–0.890	1.41	≤1100	80	[52]
Copper Slag[a]	3500	0.7–1.1	1.6	≤800	–	[25]
Copper Slag[b]	3700	1.4–1.5	2.1	≤800	–	[25]
Copper Slag-P	3600	1.18	0.8	≤800	150	[47]
Copper Slag-B	3700	0.99	1.1	≤800	150	[47]
Coal Slag	2761–3044	0.7–1.0	–	≤1000	–	[53]
Steel Slag	4204	0.63–0.89	–	≤380	–	[49]
AAM[c]	3700	1.4–1.5	2.1	≤500	70	[54]
Solar Salt	1870	1.6	0.42	≤580	625	[53]
Therminol VP1	774.2	2.41	0.09	≤400	2100	[55,56]

[a] ES-N
[b] ES-A
[c] [Fly ash/ Black Slag]

- Potential reduction of the initial acquisition costs.
- Operation at high temperatures.
- Comparable energy density with commercial solar salts.
- Valorization of industrial by-products.

This study presents experimental evaluations of axial (A-PBTES) and radial (R-PBTES) packed-bed configurations that use copper slag from Chilean foundries as the storage medium. Both rigs operate with air as the heat-transfer fluid under medium- and high-temperature conditions. The A-PBTES comprises a vertically oriented bed instrumented along the height at a fixed radius, whereas the R-PBTES is instrumented at multiple radii and elevations to resolve the two-dimensional temperature field. Both systems reverse the flow direction between charging and discharging to promote fluid–solid heat exchange. In that context, the article's main contributions are the experimentally grounded evaluation of axial and radial topologies under common boundary conditions and identical materials; the spatially resolved measurements of temperature evolution in copper-slag beds across the relevant operating range; and insights into how design topology and flow management govern thermal stratification and overall performance.

Finally, it is important to note that the present study uses fresh copper slags. While cyclic mechanical degradation (ratcheting) is a critical factor for long-term packed bed stability, it lies outside the scope of this work. Additionally, to assess the quality of thermal stratification beyond the First Law of Thermodynamics, a transient specific exergy analysis is conducted to quantify the thermodynamic performance of both bed topologies.

Apart from this introduction, the remainder of the manuscript comprises three main sections. Section 2 outlines the experimental setup, material properties, experimental procedure, and data reduction methodology. Section 3 analyzes and evaluates the axial and radial configurations, discussing temperature profiles, stored energy and exergy, thermal power evolution, asymmetry, and pressure drop. Finally, Section 4 summarizes the main conclusions of the study.

2. Materials and methods

2.1. Experimental setup

The experiments were conducted on bidirectional A-PBTES and R-PBTES prototypes, which allowed reversed flow between charging and discharging processes. Fig. 2 shows the experimental setup for both prototypes, which comprises a centrifugal fan (Sodeca model: CMP-820-2T 1.1 kW), an electric heater (Vías Importaciones SPA 35 kW 380 V), piping, and the copper slag packed bed (CSPB). Additionally, the Pyrogel XT-E insulation layer installed between the rock bed and the internal wall tank reduced the heat loss and parasitic thermal storage in the structure. The air flow was measured by using a hot wire anemometer (Kanomax model: Anemomaster 6162) – FT (flow transducer) in Fig. 2 –, and the data acquisition was executed by two dataloggers (Keysight DAQ970 A and Campbell Scientific CR1000). The test bench also included temperature (TT) and differential pressure (PT) transducers, as shown in Fig. 2. According to Table 1, a previous work [25] evaluated the copper slags properties, showing a high

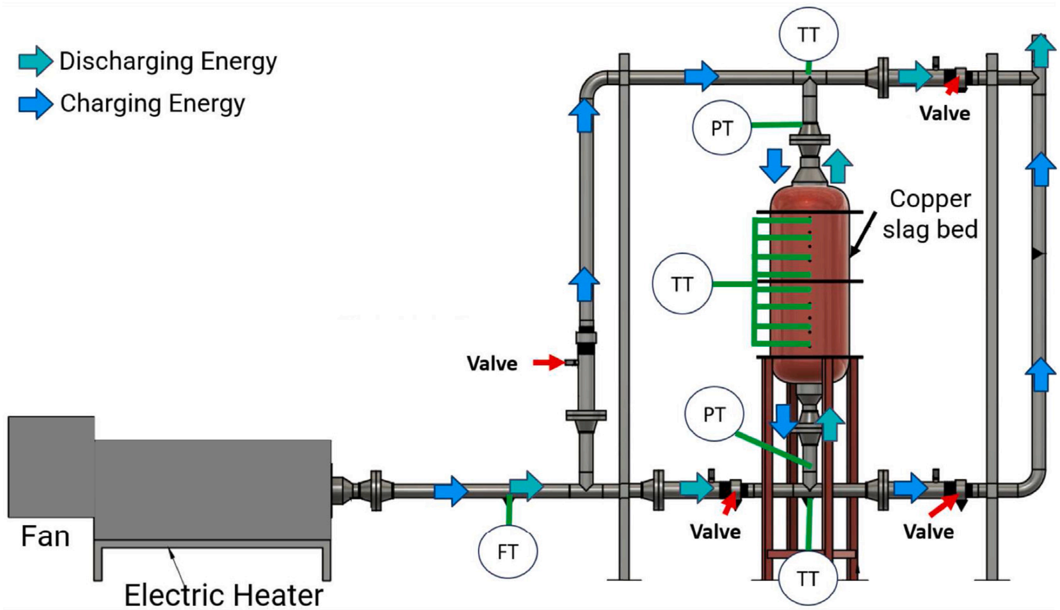


Fig. 2. Schematics of the experimental test bench.

thermal stability up to 800 °C, reporting specific heat in the range of 0.7 to 1.5 kJ kg^{-1} K^{-1}, thermal conductivity of 1.6 to 2.1 W m^{-1} K^{-1}, and density of 3500 to 3700 kg m^{-3}.

Fig. 3(a) depicts the experimental setup for the A-PBTES prototype, a cylinder that has a capacity of roughly 34 L (diameter of 280 mm and height of 551 mm). Eight type K thermocouples were installed along the vertical axis at the center of the transversal cross-section to measure in-bed temperatures, as illustrated in Fig. 3(b). The axial positions of the sensors are listed in Table 2.

Fig. 4(a) shows the R-PBTES prototype, which has a capacity of roughly 11 L (major diameter of 240 mm, minor diameter of 60 mm, and height of 256 mm). The overall rig matches the A-PBTES configuration, except for the in-bed instrumentation: eighteen type-K thermocouples were installed in the CSPB at the locations listed in Table 2 and Fig. 4(b). Such thermocouples were mounted on six steel plates placed at different circumferential angles, three thermocouples per plate (at different elevations). Two plates were positioned at the inner radius with angles of 180° and 360°, and four plates at the external radius at angles of 90°, 180°, 270°, and 360°, respectively.

For geometric consistency in the copper slag packing material, a manual selection process based on visual inspection segregated particles with a representative length scale of 20 mm to compose the packed beds. Additionally, water displacement measurements determined the A-PBTES and R-PBTES porosities to be 46% and 63%, respectively. The different values are due to the distinct arrangements of particles within the vessels since the porosity depends on the particle-to-bed size ratio [62]. Therefore, since the two prototypes differ in total bed volume, geometry (aspect ratio) and porosity, absolute stored/recovered energies are not strictly comparable as a one-to-one topology benchmark. Accordingly, the manuscript reports absolute and per-slag-mass metrics to separate geometry-driven scaling from topology-related effects.

2.2. Experimental procedure

All tests considered two charging set-point temperatures, i.e., 150 °C and 300 °C, and a measurement and recording interval of 10 s for temperatures, airflow, and pressure drop. Charging was performed from top to bottom with a time interval of either 1.5 h or 3 h, by forcing hot air through the CSPB driven by the centrifugal fan and heated by a resistive electric heater (Fig. 2). Then, during a subsequent standby phase, room air circulated through the piping, bypassing the CSPB to cool the heater and minimize the thermal energy stored in the piping. Discharging was conducted from bottom to top with room air flowing through the CSPB for the same duration as charging (1.5 h or 3 h), retrieving the stored thermal energy. The four valves depicted in Fig. 2 controlled the airflow direction during the charging and discharging phases. Flow reversal between stages was implemented to align with the thermocline developed during the charging phase (the temperature front that develops from the top towards the bottom of the CSPB). Table 3 summarizes the eight tests considered, indicating the charging temperature and duration. All experiments were duplicated on different days to assess repeatability (see Appendix A).

Table 2
Thermocouples position inside of the A-PBTES and R-PBTES.

A-PBTES		R-PBTES					
	Height [mm]		Height [mm]	radius [mm]		Height [mm]	radius [mm]
T_1	501	T_1	26	117	T_{10}	63	117
T_2	437	T_2	94	117	T_{11}	134	117
T_3	386	T_3	196	117	T_{12}	226	117
T_4	332	T_4	25	45	T_{13}	23	117
T_5	232	T_5	96	45	T_{14}	96	117
T_6	174	T_6	171	45	T_{15}	170	117
T_7	120	T_7	59	45	T_{16}	59	117
T_8	60	T_8	132	45	T_{17}	128	117
		T_9	204	45	T_{18}	231	117

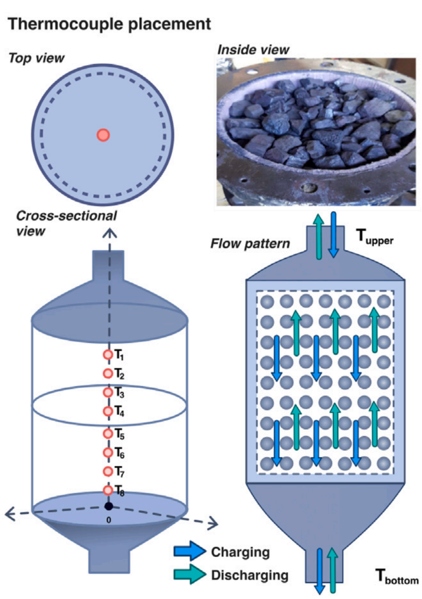
2.3. Thermal performance

Eq. (1) quantifies the instantaneous thermal power transferred to (from) the CSPB during the charging (discharging) process

$$\dot{Q}_{slag,cha} = \dot{Q}_{slag,dis} = \sum_{i}^{N}(1-\varepsilon)(\rho c)_{slag} V_i \left|\frac{T_i - T_i^0}{\Delta t}\right|, \quad (1)$$

where i indexes the N control volumes in which the system is discretized for the assessment, V_i is the control volume, ε is bed porosity, ρ_{slag} = 3700 kg m^{-3} and c_{slag} = 1.1 kJ kg^{-1} K^{-1} are the copper slag

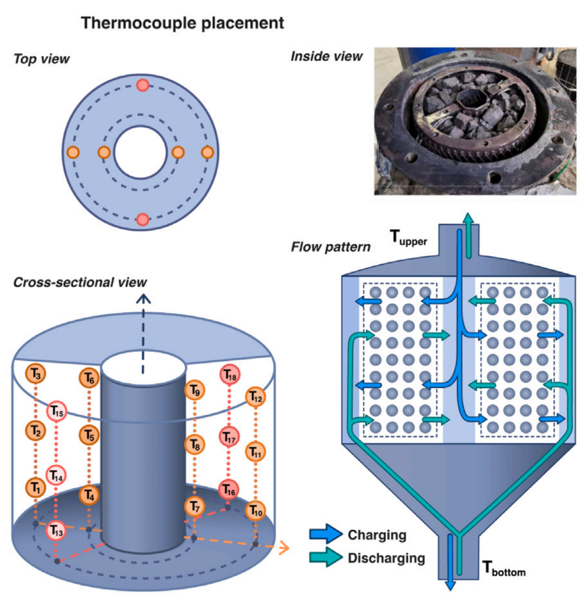
(a) A-PBTES experimental prototype.



(b) Position of the thermocouples and inside view of the A-PBTES.

Fig. 3. A-PBTES device.

(a) R-PBTES experimental prototype.



(b) Position of the thermocouples and inside view of the R-PBTES.

Fig. 4. R-PBTES device.

Table 3
Experimental tests considered.

Run	Topology	Temperature [°C]	Charging time [h]
A1	Axial	150	1.5
A2	Axial	150	3.0
A3	Axial	300	1.5
A4	Axial	300	3.0
R1	Radial	150	1.5
R2	Radial	150	3.0
R3	Radial	300	1.5
R4	Radial	300	3.0

density and specific heat, respectively, T is the measured temperature, and Δt is the sampling interval. For the temperatures, the superscript 0 indicates the previous time step, and no superscript indicates the current time step. Also, because the thermal capacity of the copper slag is much larger than that of the air, the expression accounts only for energy stored in the solid phase and neglects the gas contribution.

Considering the thermal power definition from Eqs. (1), (2) defines the round-trip thermal efficiency of the CSPB as the ratio of the discharged-charge energy over the operation cycle, i.e.,

$$\eta_{slag,round-trip} = \frac{\int_0^{t_{dis}} \dot{Q}_{slag,dis}\ dt}{\int_0^{t_{cha}} \dot{Q}_{slag,cha}\ dt}. \tag{2}$$

By disregarding the air contribution to the storage capacity in Eqs. (1), (2) isolates the performance of the solid filler and thus pinpoints the copper slag's effectiveness as a thermal storage material.

Additionally, the transient specific stored exergy of the solid phase

$$ex_{slag} = \frac{\sum_i^N (1-\varepsilon)(\rho c)_{slag} V_i \left[(T_i - T_0) - T_0 \ln\left(\frac{T_i}{T_0}\right)\right]}{m_{slag}}, \quad (3)$$

where $m_{slag} = \sum_i^N (1-\varepsilon)\rho_{slag} V_i$ is the total mass of copper slag and $T_0 = 20\,°\mathrm{C}$ is the reference temperature taken as the initial ambient temperature – temperatures in the logarithmic term are expressed in Kelvin. Eq. (3) assess the quality of the thermal stratification beyond the energy balance. This metric quantifies the useful work potential per unit mass of the copper slag relative to the dead state. Such a per-unit-of-total-mass figure of merit enables a direct comparison between the axial and radial prototypes despite their geometric differences. The analysis focuses on the non-flow exergy of the copper slag because the airflow pressure drop is not available for all charging and discharging processes.

3. Results

The results synthesize the thermal and hydraulic behavior of the A-PBTES and R-PBTES copper–slag packed beds under two inlet setpoints, i.e., 150 °C and 300 °C, and two operating windows, i.e., 1.5 h and 3 h. The discussion first analyzes the spatio–temporal temperature fields to elucidate stratification, standby losses, and discharge dynamics in the axial (Fig. 5) and the radial beds – highlighting core-to-wall gradients and azimuthal effects – using temperature profiles and maps (Figs. 7, 8, 9). Furthermore, an asymmetry index in Eq. (4) quantifies the azimuthal non-uniformities. We subsequently assess energy storage performance from instantaneous thermal power traces and integral balances, reporting $Q_{slag,cha}$, $Q_{slag,dis}$, and $\eta_{slag,round-trip}$ (Fig. 11, Table 4), and discuss the hydraulic penalty through pressure-drop measurements (Fig. 12). Unless otherwise noted, the air contribution to storage is neglected consistent with Section 2 (Eqs. (1) and (2)), and reproducibility is documented with duplicate runs in Appendix A.

3.1. Thermal behavior of the axial configuration

This subsection analyzes the formation and evolution of the thermocline in the axial copper-slag packed bed and its dependence on charging time and inlet temperature. Fig. 5 (panels a–d) displays the temporal evolution of the packed bed (air and copper slag) temperatures at eight axial positions and at the bed inlet and outlet (solid and dotted black lines, respectively). Runs A1-A4 correspond to inlet setpoints of 150 °C (A1-A2) and 300 °C (A3-A4) with charging times of 1.5 h (A1 and A3) and 3 h (A2 and A4). Two vertical dashed lines delimit the standby interval between charging and discharging. Additionally, the map shown in Fig. 6 complements Fig. 5 by facilitating the visualization of the temperature evolution over time across the entire axial domain for run A2.

Charging stage With the hot air entering the bed from the top and flowing downward, the heat transferred to the slag is stored as sensible heat, generating a stratified profile that advances downward along the flow direction. In A1 (150 °C, 1.5 h), the upper sensors rapidly approach a quasi-plateau, whereas the lower sensors remain comparatively unchanged by the end of the charging process. This behavior indicates that the heated region does not reach the bottom. Extending the charging time to 3 h (A2) promotes a deeper penetration of the heated region and a noticeable reduction of axial temperature differences near the end of charge (differences below about 5 °C). Charging at 300 °C (A3-A4) exhibits a similar qualitative behavior: the larger driving temperature enhances the initial heating rates and shortens the time for the heated region to reach intermediate depths, whereas longer charging promotes a more uniform axial profile. These trends are consistent with advection-dominated heating with progressive smoothing over time due to internal redistribution mechanisms. Additionally, the fine oscillation in the inlet temperature is due to the heater control system that uses an on/off strategy to control the air temperature at the target value. Also, a damped oscillation is present in the first thermocouple, arguably indicating that this temperature reflects the air temperature more closely than the copper slag's.

Standby stage During the standby period, the heater is turned off, and the flow bypasses the bed; nevertheless, the bed cools due to environmental losses. At an inlet setpoint of 150 °C, average temperature drops of approximately 7 °C are observed near the upper and lower ends and about 3 °C in the middle region. At 300 °C, these losses increase to roughly 20 °C (upper/lower) and 7 °C (middle). In panels (c)–(d), a slight transient rise is observed at the bottom during standby, attributable to internal heat redistribution from hotter layers once external heating ceases. Both behaviors underscore the relevance of end effects and the surface-area-to-volume penalty in small prototypes, motivating improved insulation and the minimization of metallic heat bridges for scale-up.

Discharging stage Discharge is initiated by introducing ambient air at the bottom, thereby reversing the flow with respect to charging processes, aiming to maximize the local temperature difference at the advancing front. Cooling proceeds in the reverse axial order of the sensors relative to charging, confirming persistent thermal stratification. After a 1.5 h charge (A1 and A3), the discharge curves exhibit steep initial cooling of the lowest sensors and a short period of hotter outlet air, signatures of a sharper thermocline. After a 3 h charge (A2 and A4), the outlet temperature becomes more moderate and sustained, reflecting a more uniform axial distribution at the end of charge. These observations indicate a practical trade-off: extending the charge increases the total energy stored but tends to moderate the initial discharge temperature; shorter charges preserve a sharper temperature contrast and enable higher-quality heat delivery early in discharge.

Effective inlet temperature and auxiliary inertia The effective temperature measured at the bed entrance remained below the setpoint across runs (upper black trace) despite the controlled inlet air setpoint of either 150 °C or 300 °C. This offset arises from parasitic thermal capacitance and heat losses in upstream piping and components, which absorb part of the input energy and reduce the effective temperature swing experienced by the bed. Mitigation strategies include minimizing dead volume between heater and bed, shortening and insulating hot runs, thermally isolating valves and flanges, and, when applicable, preheating critical sections to reduce the initial energy sink.

Overall assessment of A-PBTES Overall, the A-PBTES exhibits stable, repeatable stratification across the tested conditions. Higher inlet temperature accelerates the establishment of the heated region, whereas longer charging deepens penetration and yields a more uniform axial distribution by the end of charge. These findings support the suitability of copper slags as a low-cost filler capable of sustaining stratified operation across duplicated runs under the tested conditions, and they clarify the operational trade-off between maximizing stored energy and preserving high initial discharge temperatures.

3.2 Thermal behavior of the radial configuration

This subsection examines the spatio–temporal temperature evolution of the radial packed bed and its dependence on inlet temperature and operating time. Fig. 7 (panels a–d) presents the temperature histories recorded at multiple heights and at two radii (inner and outer annulus), together with the inlet and outlet traces. Runs R1–R4 correspond to inlet setpoints of 150 °C (R1 and R2) and 300 °C (R3 and R4), with charging times of 1.5 h (R1 and R3) and 3 h (R2 and R4). A practical feature of this configuration is that the heater control does not perturb the copper slag–air interface temperature at the bed inlet; consequently, the thermocouples at the internal radius reliably track the effective inlet condition employed in the analysis. Additional temperature measurements for the radial packed bed are provided in

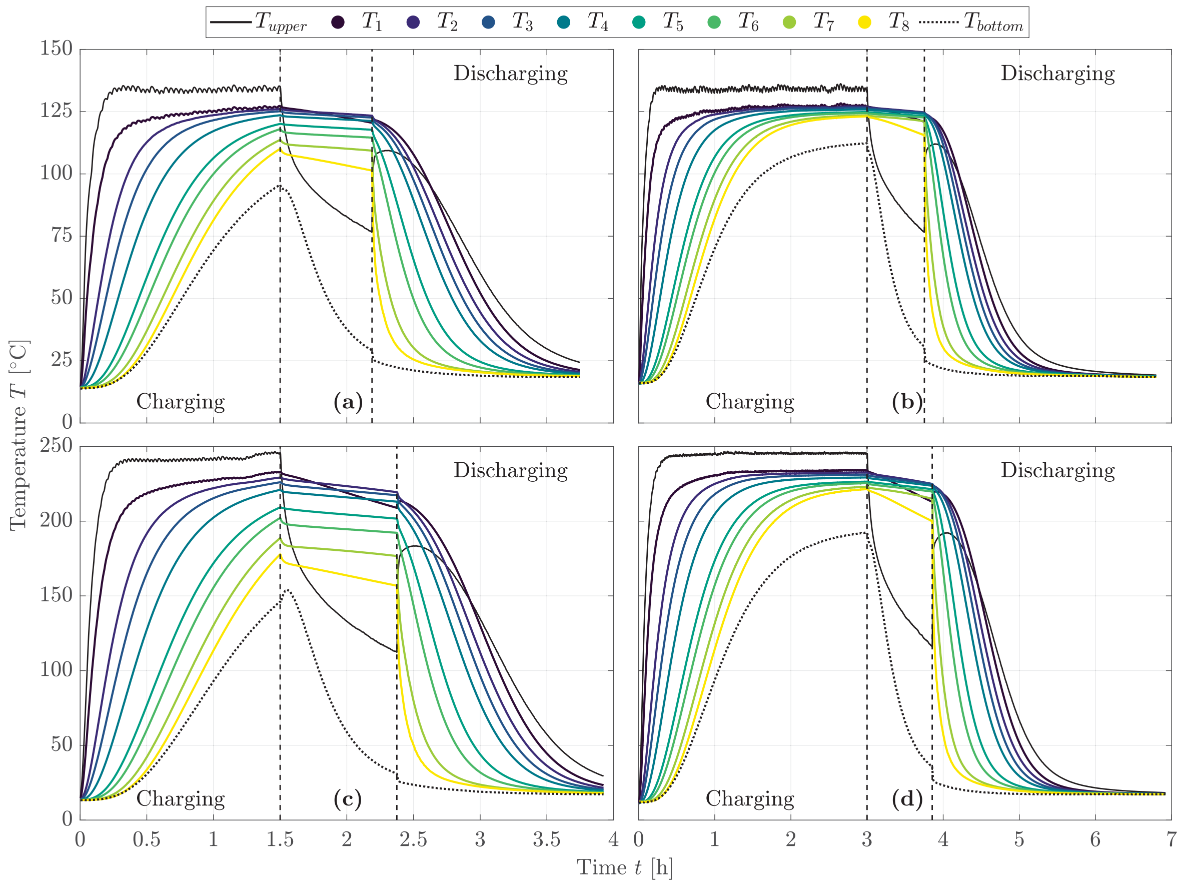


Fig. 5. Temperature profiles along the time for the A-PBTES device: (a) Run A1, (b) Run A2, (c) Run A3, and (d) Run A4.

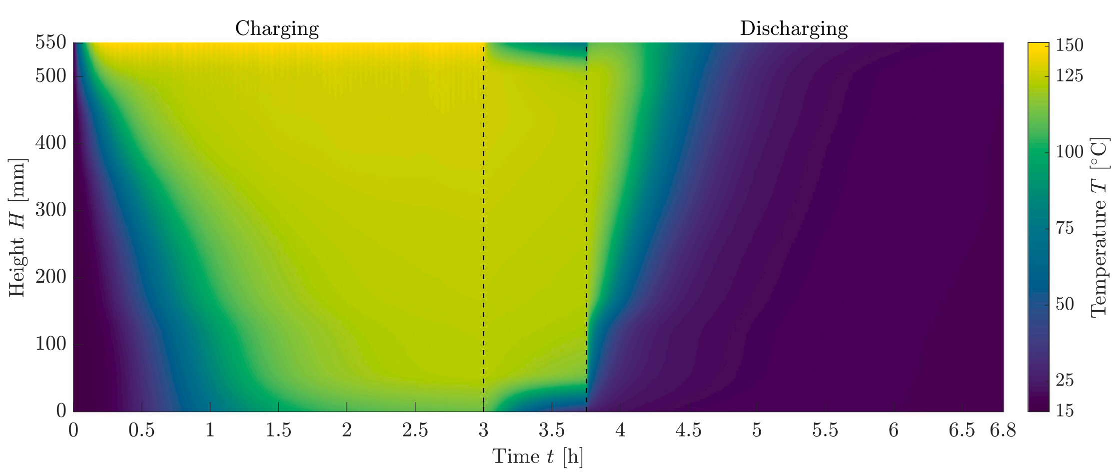


Fig. 6. Temperature evolution map for the Run A2 of the A-PBTES device.

Appendix B. Additionally, Figs. 8 and 9 complement Fig. 7 by showing temperature maps during charging and discharging processes of the R-PBTES for the run R2.

Charging stage When hot air is introduced, the heated region forms near the inner radius and propagates radially outward while also spreading axially. Across all runs, inner-radius sensors radius warm earlier and reach higher temperatures than outer-radius counterparts, confirming the expected core-to-wall radial gradient. At the lower setpoint (150 °C, R1–R2), the propagation of the heated region is slower and axial non-uniformities are more visible at the beginning of the charging phase. Extending the charging time to 3 h (R2) drives deeper penetration radially and axially, reducing inter-sensor differences. At the higher setpoint (300 °C, R3–R4), the larger driving temperature accelerates the initial heating of the inner radius and shortens the time for the heated region to reach intermediate radii, while longer charging (R4) again yields a more uniform profile by the end of the stage. These trends are consistent with radially outward advection governed by local velocity distribution and progressive internal redistribution over time.

Standby stage During standby, the flow bypasses the bed, and the heater is turned off; yet the bed undergoes measurable cooling due

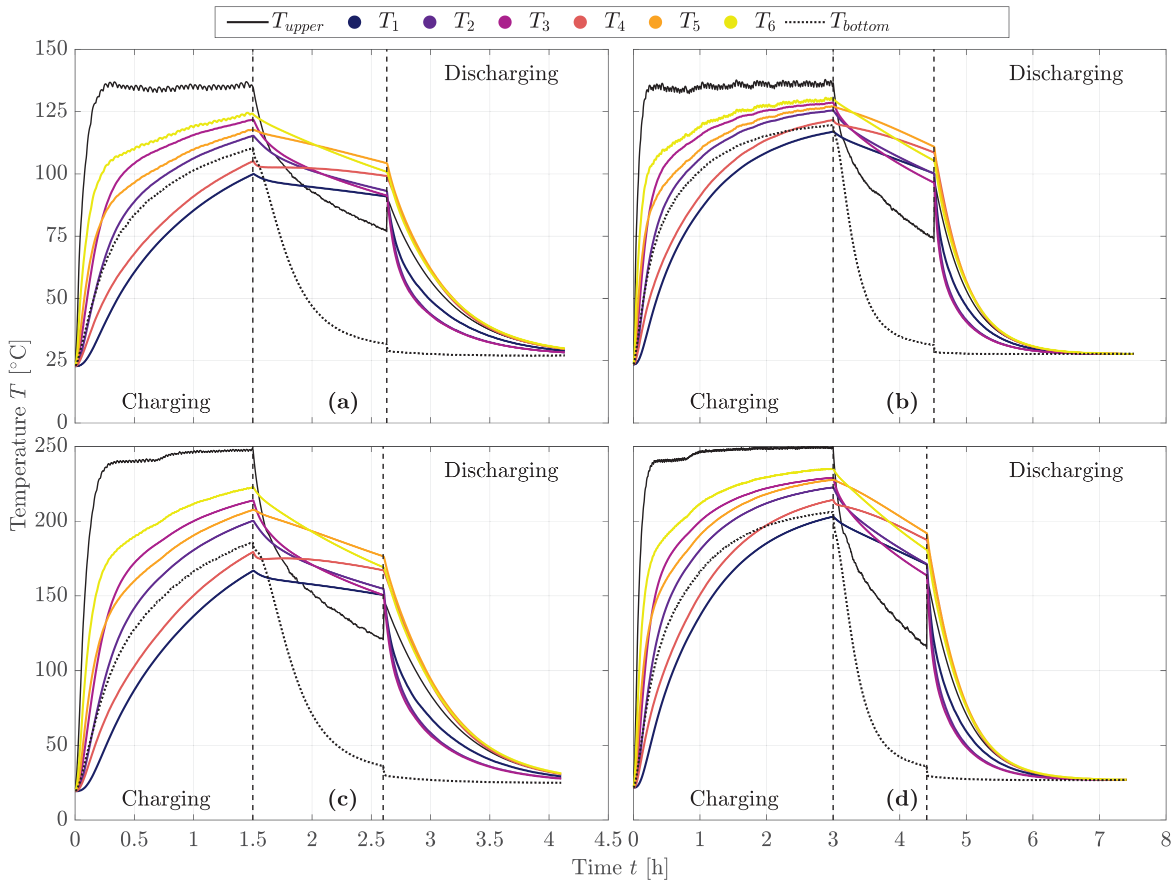


Fig. 7. Temperature profiles along time for the R-PBTES device: (a) Run R1, (b) Run R2, (c) Run R3, and (d) Run R4. Inner- and outer-radius thermocouples at multiple heights are shown along with inlet and outlet traces; vertical dashed lines mark the standby interval.

to environmental losses. The cooling pattern is not uniform: the outer radius cools faster than the inner radius at matching heights, reflecting greater lateral heat exchange with the surroundings. A slight temperature redistribution occurs immediately after heating ceases at mid-heights, indicative of internal rebalancing from hotter central regions towards cooler peripheral zones. As in the axial case, losses are larger at 300 °C, underscoring the relevance of wall insulation and end sealing to minimize parasitic heat loss.

Discharging stage Discharge proceeds by reversing the flow relative to charging. The cooling sequence of sensors confirms the preservation of the previously established radial and axial distributions: at a given height, outer-radius sensors cool sooner, and inner-radius sensors maintain elevated temperatures for longer. After 1.5 h of charge (R1 and R3), a pronounced radial contrast produces a short interval of higher outlet temperature followed by a rapid decline as the cooler peripheral zones dominate the flow path. After 3 h of charge (R2 and R4), discharge becomes more sustained and moderate, consistent with the more uniform field achieved at the end of charging. These results highlight the operational trade-off between enabling rapid power delivery early in discharge (strong contrast) and obtaining a steadier, longer release (greater uniformity).

Radial distribution and self-insulation The temperature profiles in Fig. 7 and the temperature maps during charging/discharging in Figs. 8 and 9 consistently indicate higher temperatures around the bed mid-height and closer to the inner radius. The mid-height region benefits from lower end losses, while the central annulus experiences reduced lateral dissipation and acts as a quasi self-insulated core. The resulting core-to-wall temperature difference is advantageous for reducing sidewall losses. The same feature, however, may promote non-uniform utilization of the storage volume in case of flow distributions are not properly managed.

Asymmetry analysis An asymmetry index I_a quantifies azimuthal non-uniformity by comparing temperatures at diametrically opposed circumferential positions. The proposed index is expressed as follows:

$$I_a = \sqrt{\frac{\sum_{i=1}^{N_i}\sum_{j=1}^{N_j}\left(T_{i,j} - T_{i,-j}\right)^2}{N_i N_j}}. \tag{4}$$

where, i and j identify axial and circumferential nodes, respectively. Fig. 10 shows a representative case in which I_a peaks at the onset of both charging and discharging and then decreases progressively over time. Across runs, charging exhibits the largest asymmetry, with maxima around 11 %, while discharging remains between 5 % and 7 %. Greater asymmetry is observed at lower inlet temperature or under extended operation, pointing to the combined effect of inlet/plenum distribution and geometric tolerances. From a design standpoint, these findings support using symmetric inlet diffusers, perforated plates, or graded-porosity zones to mitigate preferential flow paths and enhance uniform bed utilization.

Overall assessment of R-PBTES The R-PBTES maintains a hot central region with radial and axial gradients that evolve predictably with inlet temperature and operating time. This configuration leverages an inherent self-insulation effect (hot core, cooler periphery) that reduces lateral heat losses. At the same time, its early-time temperature contrast enables rapid power delivery at the start of discharge. Measurable azimuthal asymmetries, however, emphasize the importance of flow-distribution hardware to avoid dead zones and improve the effective

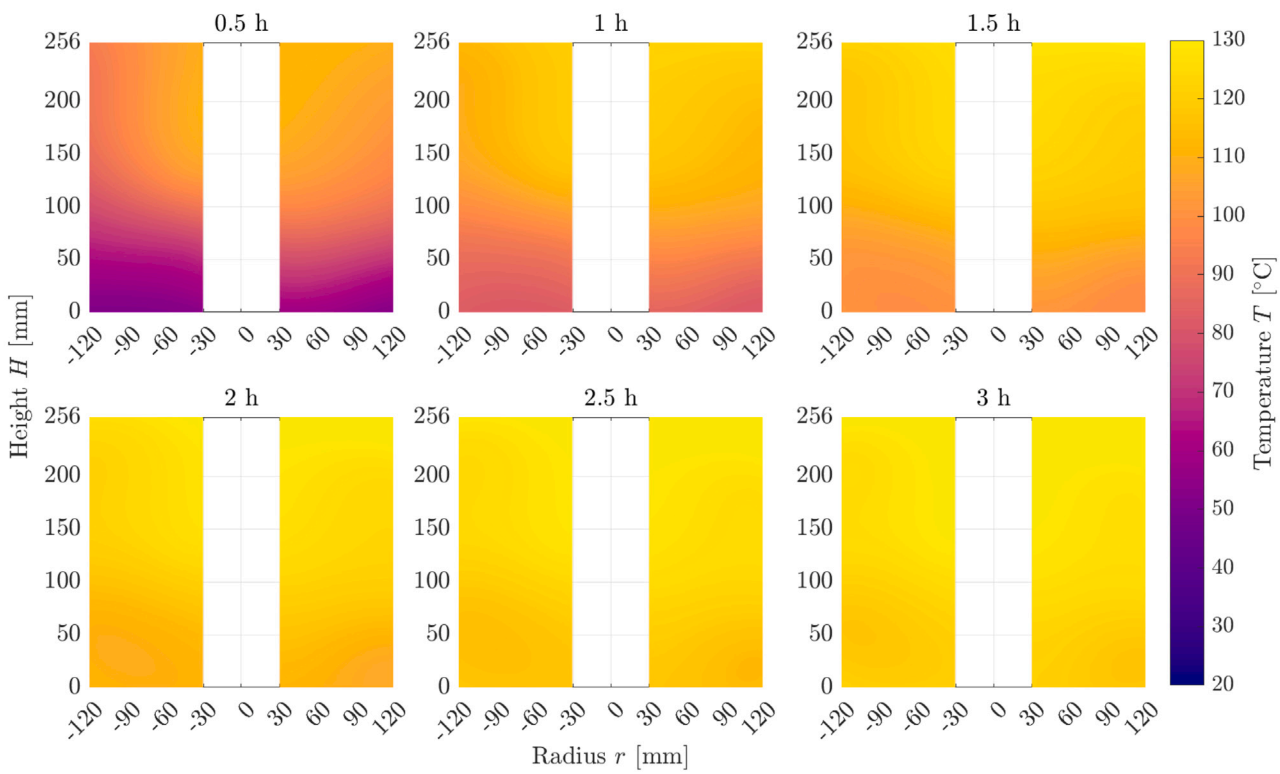


Fig. 8. Temperature evolution maps during *charging* for the R-PBTES device for run R2. Warmer colors develop first near the inner radius and mid-height, propagating radially outward and spreading axially.

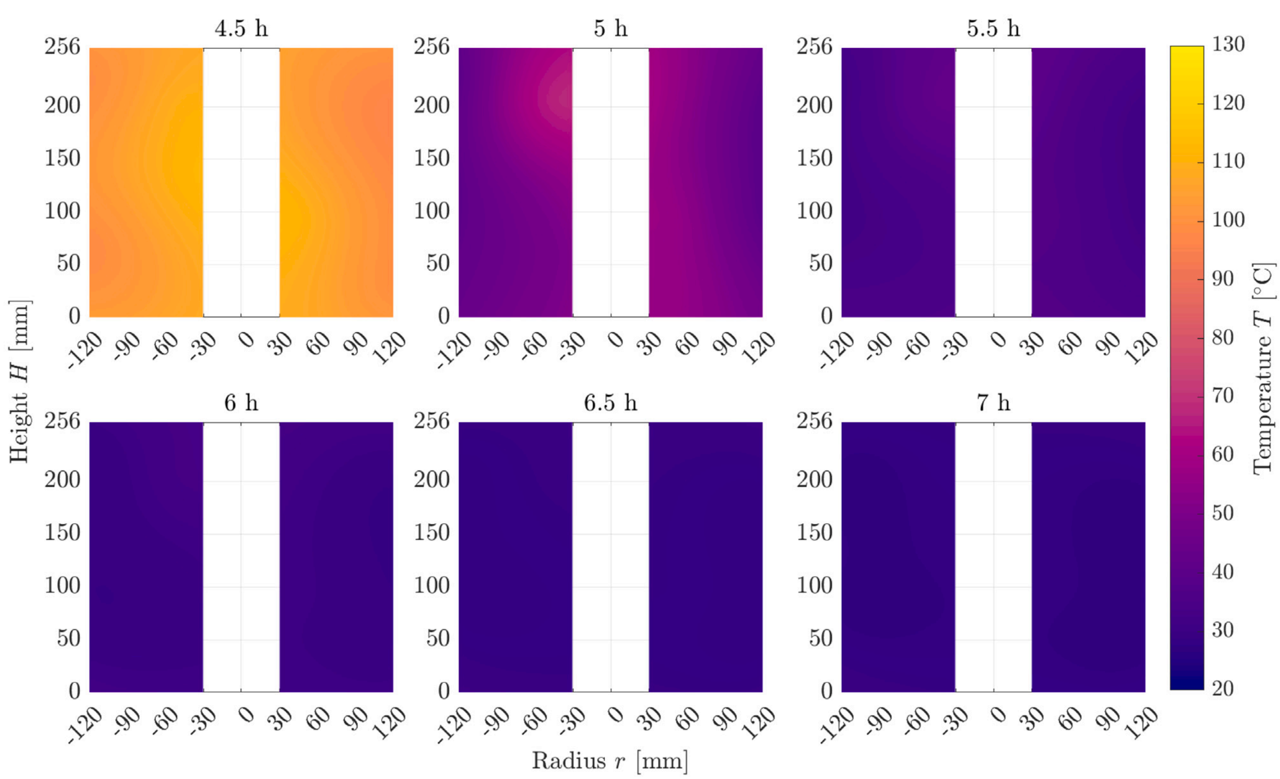


Fig. 9. Temperature evolution maps during *discharging* for the R-PBTES device for run R2. Cooling starts earlier at the outer radius, while the inner radius retains higher temperatures longer, evidencing preservation of prior stratification.

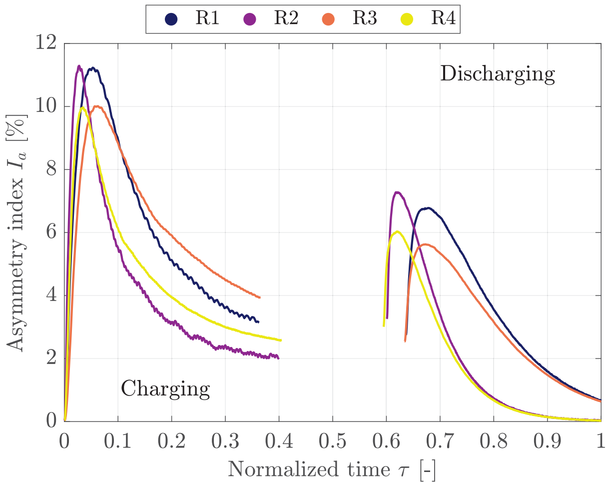


Fig. 10. Asymmetry index I_a for the R-PBTES (example for run R2). Peak asymmetry appears at the onset of both charging and discharging and decreases with time as the temperature field homogenizes.

Table 4
Thermal performance of the A-PBTES and R-PBTES.

Run	$Q_{\text{slag,cha}}$ [kW h]	$Q_{\text{slag,dis}}$ [kW h]	$\eta_{\text{slag,round-trip}}$ [%]
A1	1.52	1.40	91.9
A2	1.56	1.49	95.5
A3	2.84	2.62	92.3
A4	3.08	2.91	94.4
R1	0.42	0.31	73.5
R2	0.47	0.35	74.1
R3	0.82	0.60	73.8
R4	0.92	0.69	74.8

use of the storage volume. Overall, the observed behavior aligns with the expected physics of radial packed beds and provides actionable guidance for scaling the configuration towards higher capacities with controlled non-uniformities.

3.3 Thermal-hydraulic performance assessment

This subsection evaluates the instantaneous thermal power, the stored and recovered energy, and the round-trip efficiency of the copper slag packed beds, integrating the time measurements and the per-run energy balances (i.e., Eqs. (1) and (2)). Unless otherwise stated, the energy stored in air is considered negligible relative to the solids, consistent with the methodology used to compute $Q_{\text{slag,cha}}$, $Q_{\text{slag,dis}}$, and $\eta_{\text{slag,round-trip}}$.

Instantaneous thermal power Fig. 11 presents the evolution of the thermal power for the A-PBTES (left) and R-PBTES (right). In all cases, the power peaks early during charging when the temperature difference between the hot air and the colder bed is largest, and then decays as the bed warms and the driving ΔT reduces. The discharging process exhibits a mirror behavior: the power is initially high due to the strong ΔT between the colder inlet air and the hotter solids, decreasing as the bed cools. The system does not reach steady state within 1.5 h of charging, which is consistent with the persistence of stratification observed in Section 3.1. In contrast, the discharging periods approach quasi-steady conditions in all runs, exhibiting higher heat transfer rates at the beginning of the stage. Additionally, the faster early-time dynamics identified for R-PBTES in Section 3.2 are consistent with an initially larger power burst followed by a quicker decline, owing to the pronounced core-to-wall contrast and preferential flow through the central annulus.

Stored and recovered energy; round-trip efficiency Table 4 summarizes the per-run energy balances. The A-PBTES stores and recovers more energy than the R-PBTES, achieving a higher round-trip efficiency across operating conditions. Aggregating the results, the axial bed reaches 2.2 ± 0.7kW h stored and 2.1 ± 0.7kW h recovered, with $\eta_{slag,round-trip}$= 93.5 ± 1.5%, whereas the radial bed reaches 0.7 ± 0.2kW h stored and 0.5 ± 0.2kW h recovered, with $\eta_{slag,round-trip}$=73.9 ± 0.7%.

These results show that, in the present prototypes, the axial unit achieved higher absolute stored/recovered energy and round-trip efficiency under the tested conditions. However, because the two units are not geometrically matched (bed volume/geometry and porosity differ), these results should be interpreted as a prototype-level comparison rather than an intrinsic one-to-one topology ranking. The observed differences may be associated with (i) a longer residence time, (ii) a more uniform utilization of the storage volume during both charging and discharging, and (iii) a reduced sensitivity to non-uniform flow distribution. In the radial configuration, the intrinsic core-to-wall gradients and the inlet plenum distribution can promote preferential pathways that bypass parts of the bed (dead zones), limiting effective storage and reducing the recovered energy fraction. This effect aligns with the azimuthal asymmetry trends in Section 3.2. Also, considering inlet temperature oscillation discussed in Section 3.1, the metrics reported in Table 4 disregard the thermal capacities of the first and last nodes.

It is essential to acknowledge the limitations regarding local flow hydrodynamics. Although packed beds are known to experience flow channeling (particularly near the walls in axial configurations) and stagnation zones (in the corners of radial configurations), the discrete nature of the thermocouple grid used in this study does not allow for the required spatial resolution to capture these local phenomena accurately. Determining these dead zones would require further exhaustive experimental measurements and detailed computational fluid dynamics (CFD) simulations. Therefore, one should interpret the temperature evolution maps presented by Figs. 6, 8, and 9 are macroscopic representations of the beds thermal responses.

Operational pressure drop Fig. 12 reports the hydraulic behavior during the charging phase. In A-PBTES, the measured pressure drop increases at higher air temperatures, as the reduced gas density leads to higher velocities and enhanced turbulence for a given blower setting. In R-PBTES, the pressure drop remains comparatively uniform, reflecting a more even distribution of flow paths at the bed scale. From a systems perspective, lower pressure drop translates into reduced fan power and, therefore, a smaller auxiliary energy penalty. While $\eta_{\text{slag,round-trip}}$ in Table 4 isolates the storage medium, integrating hydraulic auxiliaries at the plant level further accentuates the operational advantage of configurations with lower pressure losses (notably the radial topology) in applications prioritizing fast power delivery. Conversely, when maximizing recovered energy and efficiency is paramount, the axial prototype remains preferable given its consistently higher $Q_{\text{slag,dis}}$ and $\eta_{\text{slag,round-trip}}$ across the tested conditions.

Exergetic assessment While $\eta_{\text{slag,round-trip}}$ focuses on the quantity of thermal power recovered, it treats all thermal energy stored equally, regardless of temperature. Fig. 13 presents the evolution of the specific stored exergy in the copper slag over time for both topologies across the experimental runs. This metric highlights the quality of the thermal stratification: for a fixed amount of stored energy, hence, a sharper thermocline (higher average temperature difference relative to T_0) yields higher exergy. Consistent with the temperature profiles, the A-PBTES configuration (left) exhibits superior exergy storage performance compared to the R-PBTES (right) in all tested cases. Specifically, the peak specific exergy in the axial bed was between 1.6 % and 14.5 % higher than in the radial bed. The discrepancy is most pronounced in runs A1/R1 and A3/R3 (shorter charges), where the axial bed maintains a highly stratified plug-flow-like thermal front. In contrast, the radial topology suffers from inherent thermal dispersion as the flow area increases along the radius, which promotes internal entropy generation

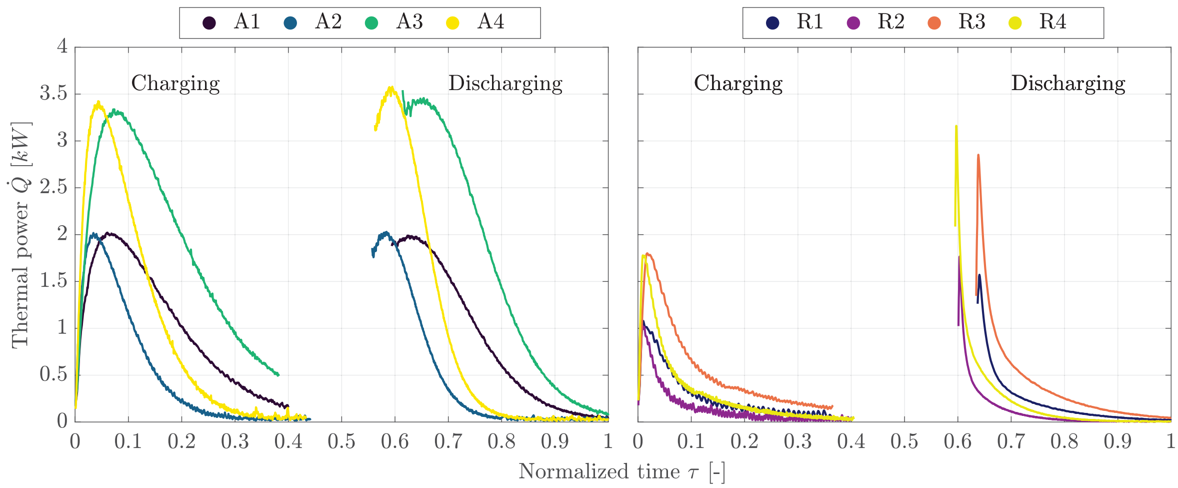


Fig. 11. Thermal power over time of the A-PBTES (left) and R-PBTES (right) devices. Power peaks at the beginning of each stage (charging/discharging) due to the larger temperature difference and decays as the bed approaches the stage end.

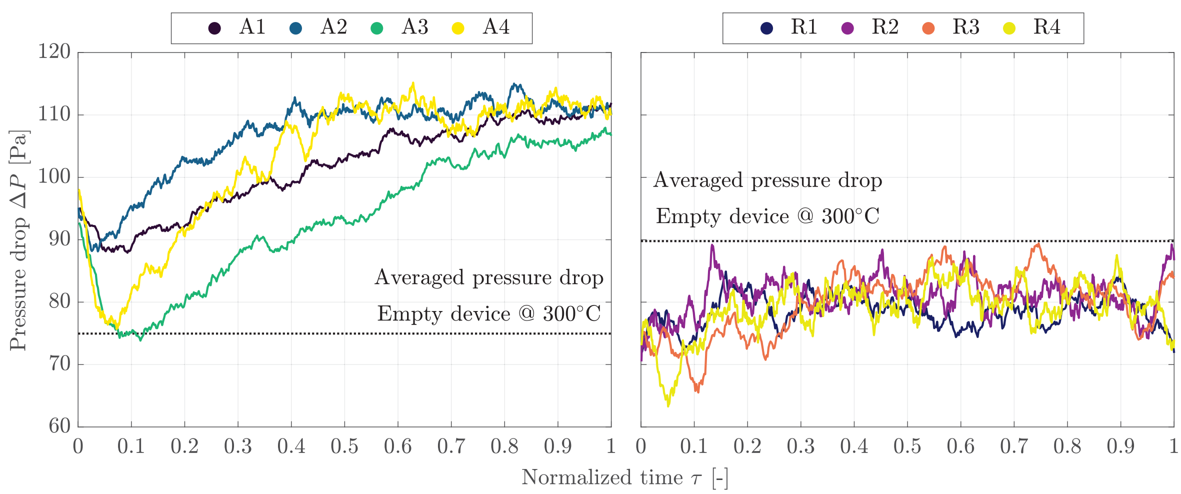


Fig. 12. Pressure drop of the A-PBTES (left) and R-PBTES (right) devices during the charging process. The axial configuration exhibits higher and temperature-dependent pressure drops, whereas the radial configuration shows a more uniform behavior.

and degrades the sharpness of the thermocline. In this sense, a trade-off appears to exist: although the R-PBTES achieved a lower pressure drop during the charge, the mixing effects within the expanding flow path slightly reduce the work potential of the stored energy.

It is worth noting that while the A-PBTES exhibits higher thermal efficiency due to better stratification, the R-PBTES operates with lower pressure drop. In a commercial-scale system, the reduction in parasitic pumping power possibly offered by the radial topology could offset its thermal disadvantages, potentially resulting in a higher overall system efficiency. However, quantifying this trade-off requires continuous pressure logging and is specific to the scale of the plant, thus remaining outside the scope of this material characterization study.

4 Conclusions

This work provided a side-by-side experimental assessment of A-PBTES and R-PBTES copper-slag packed beds for sensible heat storage, reporting the spatio–temporal temperature fields and their implications for energy recovery and hydraulics. In the axial topology, the temperature profiles confirmed the formation and preservation of a stratified region in the flow direction throughout charge, standby, and discharge (Section 3.1). Stratification was stronger after 1.5 h charges –favoring higher initial discharge temperatures – whereas 3 h charges promoted deeper penetration of the heated region and more uniform axial profiles at the end of charge. In the radial topology, the temperature maps revealed a hot central core with radial and axial gradients that evolve predictably with inlet temperature and operating time, together with a self-insulation effect that limits lateral losses but requires careful flow distribution to avoid dead zones (Section 3.2). Across both devices, standby losses increased with the inlet setpoint, highlighting the importance of end and wall insulation in scale-up.

Regarding energy balances, the axial prototype achieved higher recovered energy and $\eta_{\mathrm{slag,round-trip}}$ under the tested conditions. Nevertheless, since the two prototypes are not geometrically matched (bed volume/geometry and porosity differ), these outcomes should be interpreted as a prototype-level comparison rather than a strict one-to-one topology ranking. In the present rig, the axial configuration likely benefits from a longer residence time and a more uniform utilization of the storage volume. In contrast, the radial layout can be more susceptible to preferential pathways that bypass portions of the bed, in line with the azimuthal asymmetry trends (peak 11 % at the onset of charging and 5 % to 7 % during discharging) quantified by the asymmetry index in Section 3.2. Notably, the choice of charging duration introduces an operational trade-off: extending the charge from 1.5 h to

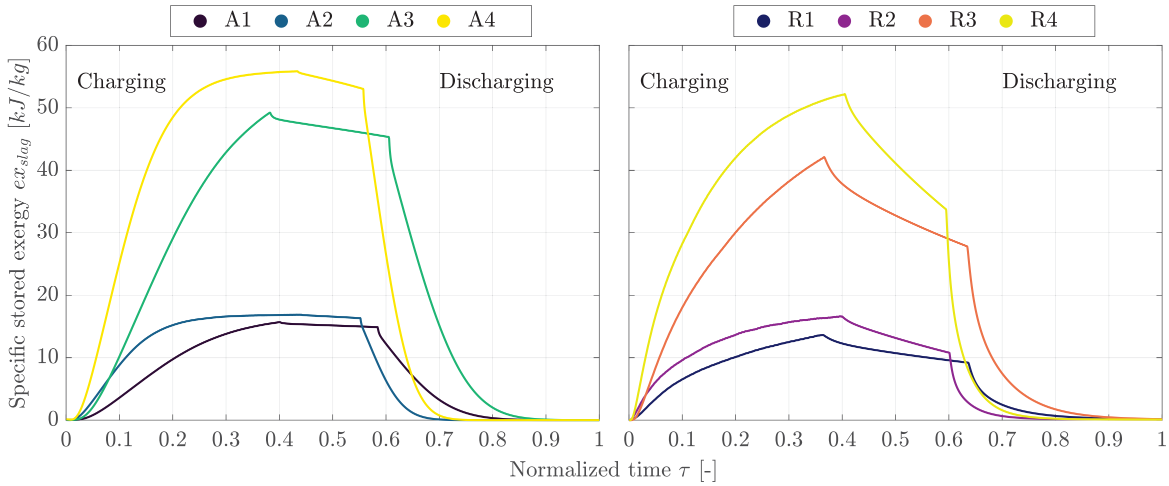


Fig. 13. Specific stored exergy of the copper slag over time of the A-PBTES (left) and R-PBTES (right) devices. The axial topology consistently achieves higher exergy density, indicating superior stratification quality.

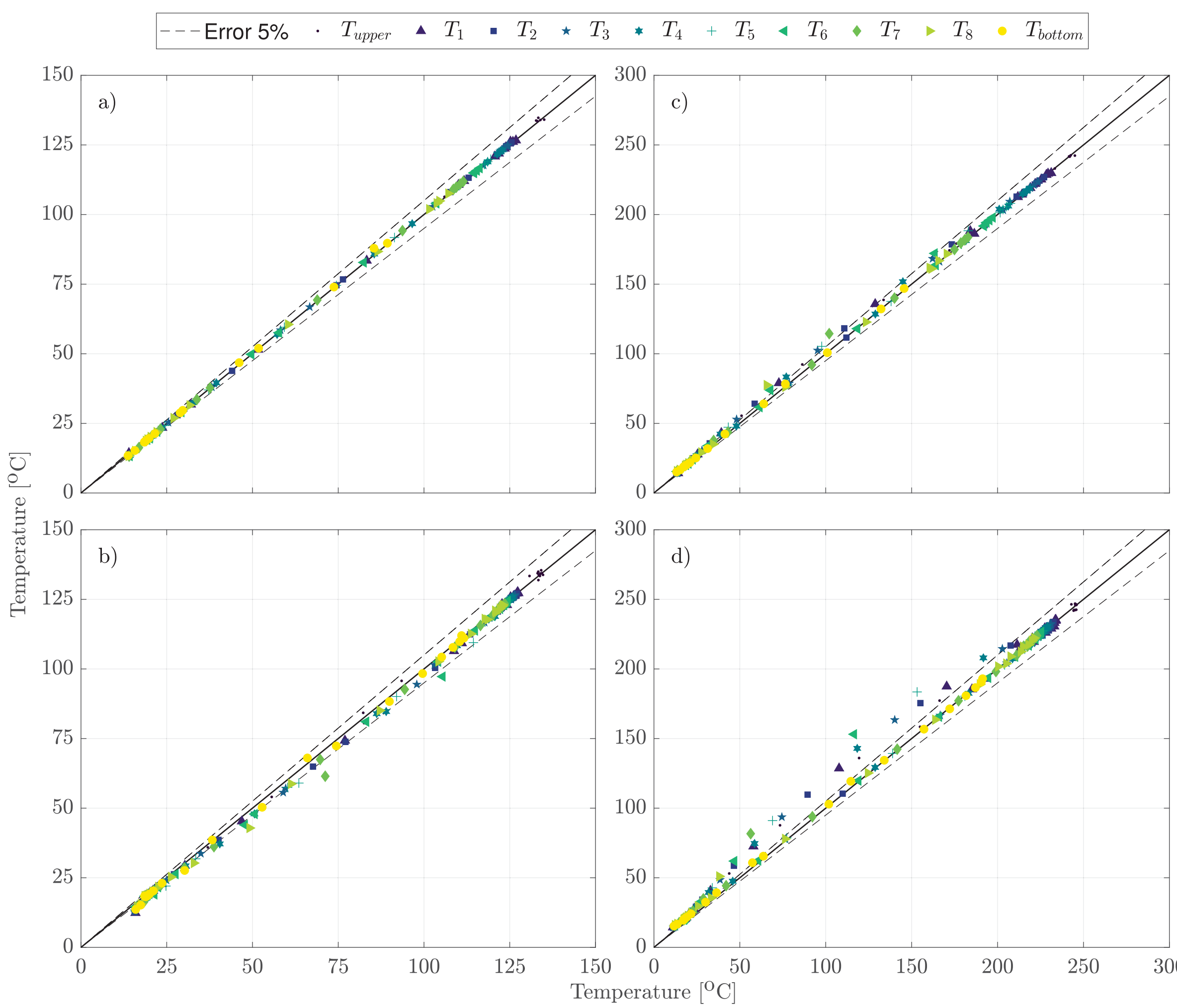


Fig. A1. Comparison among the temperature profiles for the experimental runs and their duplicated: (a) Run A1, (b) Run A2, (c) Run A3, and (d) Run A4.

3 h increases stored energy but moderates the initial discharge temperature, whereas shorter charges preserve a stronger contrast beneficial for early, higher-quality heat delivery.

The thermal power profiles provided complementary insight into the temporal dynamics. In both configurations, the instantaneous thermal power peaked at the beginning of each stage when the driving temperature difference was largest and decayed as the stage progressed (Fig. 11). The radial bed delivered a more impulsive response – storing and releasing a larger fraction of its energy in the first quarter of the stage – while the axial bed exhibited a smoother and more sustained release, consistent with its higher recovered energy. These behaviors suggest distinct application niches: R-PBTES is attractive for short, power-dense transients or peaking support, whereas A-PBTES is preferable when the objective is to maximize energy recovery and deliver steadier heat to downstream processes.

From a hydraulic standpoint, pressure-drop measurements during charging (Fig. 12) showed that the axial arrangement exhibits higher and temperature-dependent losses, attributable to increased gas velocity and turbulence at elevated air temperatures. The radial arrangement presented a more uniform pressure drop, reflecting a more even macroscopic flow distribution. These differences translate into distinct operational penalties when internalizing auxiliary consumption at the plant level.

Overall, the results validate copper slags as a low-cost, industrially sourced filler for both axial and radial packed beds. The combined evidence – temperature fields, energy balances, power profiles, and hydraulic penalties –provides actionable guidance for configuration selection and operating policies: (i) axial layouts for high-efficiency energy recovery and steadier delivery; (ii) radial layouts for rapid power dispatch under reduced auxiliary loads; and (iii) careful management of charging time, inlet temperature, and insulation to balance stored energy, discharge quality, and standby losses. Additionally, the specific stored exergy analysis demonstrated that the axial topology preserves thermodynamic potential more effectively than the radial counterpart. The axial bed achieved peak specific exergy values 1.6% to 14.5% higher than the radial bed, confirming that the variable flow area in the radial design induces greater thermal dispersion, degrading the thermocline sharpness and increasing exergy destruction compared to the plug-flow behavior of the axial configuration

Perspectives and future work Building on these results, the investigation identifies the following priorities: (i) scale-up studies addressing end/wall insulation and minimizing metallic heat bridges to curb standby losses; (ii) flow-distribution hardware (symmetric diffusers, perforated plates, graded porosity) to mitigate azimuthal non-uniformities and dead zones in R-PBTES; (iii) multi-cycle durability of copper slags (granulometry, sintering, fines generation, oxidation) and long-term performance drift; (iv) high-fidelity CFD modeling of porous media validated against the present dataset to resolve non-uniform flow distributions and predict operating maps; (v) plant-level techno-economic assessment that internalizes auxiliary power for fans and control hardware, and compares axial vs. radial layouts under realistic duty cycles; (vi) environmental and circular-economy assessments of copper-slag sourcing and leachate behavior to de-risk large-scale deployment; (vii) full-system techno-economic optimization that weighs the hydraulic benefits of the radial design against its thermal stratification penalties; and (viii) investigation into the long-term mechanical stability of the slag under thermal cycling.

CRediT authorship contribution statement

Felipe G. Battisti: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ignacio Calderón-Vásquez:** Writing – review & editing, Validation, Methodology, Investigation, Conceptualization. **Jose Ortega:** Methodology, Investigation, Formal analysis, Conceptualization. **Ian Wolde:** Methodology, Investigation, Conceptualization. **Valentina Segovia:** Methodology, Investigation, Conceptualization. **Raimundo Claren:** Methodology, Investigation, Conceptualization. **Ignacio Arias:** Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Nicolás Pailahueque:** Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Rodrigo A. Escobar:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization. **José M. Cardemil:** Writing – review & editing, Supervision, Resources, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the financial support from projects ANID/FONDEF/ID22I10200, ANID/FONDECYT/1231186, ANID/CENTRO DE INTERES NACIONAL/CIN250043 “Solar Energy Research Center” SERC Chile, and Project supported by the Fund for Scientific and Technological Equipment, year 2024, code LE24-03, Universidad Tecnológica Metropolitana. I. Calderón-Vásquez acknowledges the PhD. Scholarship ANID PFCHA/Doctorado Nacional/2021–21210778, I. Wolde acknowledges the PhD. Scholarship ANID PFCHA/Doctorado Nacional/2021-21211849, I. Arias acknowledges the PhD. Scholarship ANID PFCHA/Doctorado Nacional/2021-21210053 and V. Segovia acknowledges the PhD. Scholarship ANID PFCHA/ Doctorado Nacional/ 2023-21232013.

Appendix A. Experiment repetitiveness

Fig. A1 summarizes the reproducibility of the axial-bed experiments by comparing, for each run, the temperatures recorded in a reference experiment against those measured in its duplicate at the same thermocouple locations. The abscissa corresponds to the instantaneous temperatures of the original experiment, and the ordinate corresponds to the temperatures in the duplicate; hence, perfect reproducibility would place all points on the $y = x$ line. The dataset aggregates all operating stages (charging, standby, and discharging) and all instrumented heights, thereby providing a stringent, time-resolved consistency check across the full thermal cycle.

To quantify repeatability, we computed the RMSE between paired temperature samples,

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(T_i^{(1)} - T_i^{(2)}\right)^2}, \tag{5}$$

where N is the total number of paired measurements and $T_i^{(1)}$ and $T_i^{(2)}$ denote the i-th temperature sample in the original and duplicate experiment, respectively. We also evaluated per-sensor RMSE values to identify location-dependent effects (e.g., end losses near boundaries). Across all duplicates, the RMSE lies between 0.44 °C and 6.02 °C, indicating high reproducibility relative to the temperature swing imposed in the tests.

A qualitative inspection of Fig. A1 shows a tight clustering around the identity line, with deviations increasing slightly in time windows characterized by large thermal gradients (end of charge and onset of discharge). This behavior is consistent with the combined influence of (i) slight differences in the effective inlet temperature owing to auxiliary thermal inertia upstream of the bed, (ii) unavoidable variability in blower operation and inlet plenum flow distribution, and (iii) enhanced sensitivity to end losses at the highest absolute temperatures.

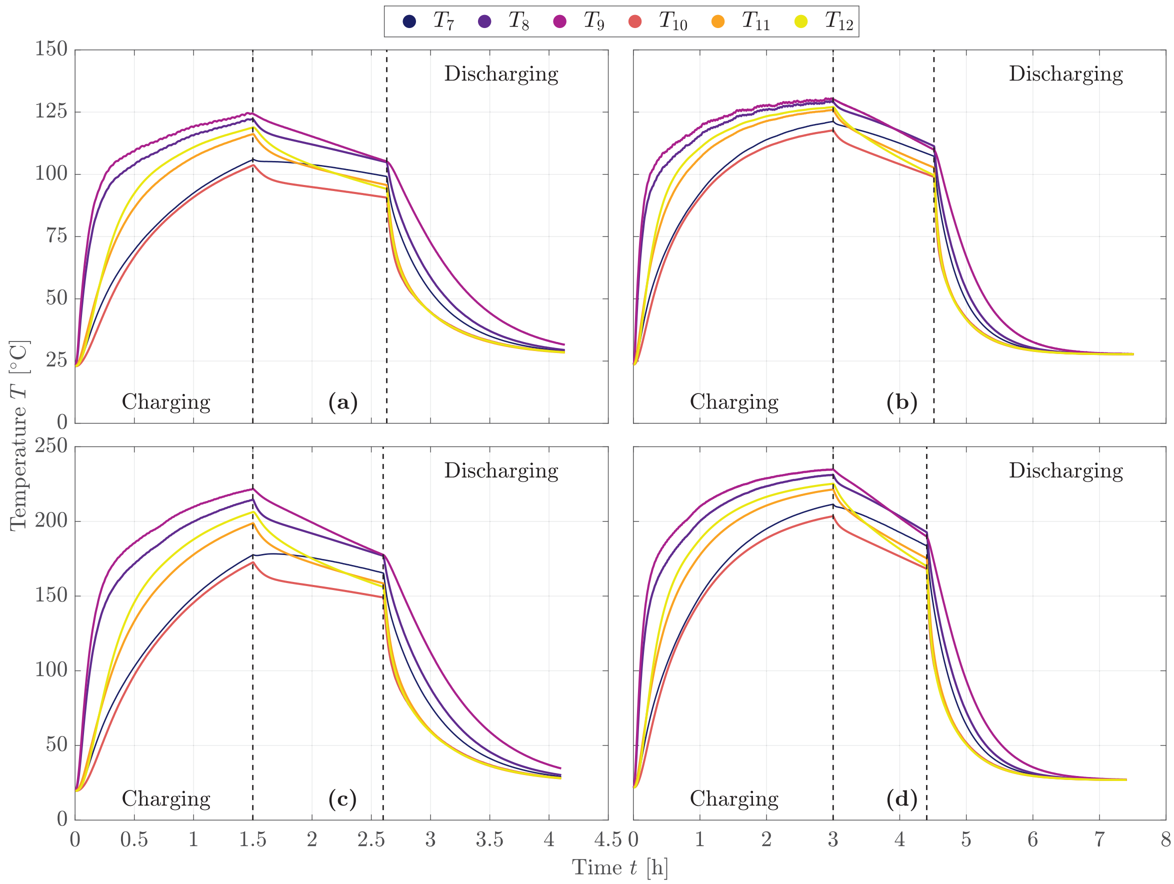


Fig. B1. Temperature profiles along the time for the R-PBTES device: (a) Run R1, (b) Run R2, (c) Run R3, and (d) Run R4.

Importantly, no systematic bias (offset) is apparent across the range of temperatures considered, which supports the absence of long-term drift in instrumentation or boundary conditions between duplicates.

From an application standpoint, the observed repeatability ensures that integral performance metrics – stored/recovered energy and round-trip efficiency – are robust to experimental variability. A normalized RMSE can explicitly relate temperature scatter to the imposed temperature swing, nRMSE $= \mathrm{RMSE}/(T_{\mathrm{in,eff}} - T_{\mathrm{amb}})$, facilitating comparisons across operating points. Future datasets may also include complementary reproducibility diagnostics (e.g., Bland–Altman plots and per-stage RMSE) to resolve further whether discrepancies concentrate in specific phases or locations; however, the present analysis already demonstrates that experimental uncertainty is small compared with the thermal signals of interest and does not alter the conclusions drawn in Section 3.

Appendix B. Temperature profiles in radial layout

This appendix complements Section 3.2 by providing the remaining temperature measurements recorded in the radial packed bed not displayed in the main text. The traces correspond to six thermocouples per panel, arranged along a diametral plane at two radii (inner and outer annulus) and three heights (bottom, mid-height, top) – Table 2 provides the exact coordinates. Each figure – Figs. B1 and B2 – gathers the four experimental runs (R1–R4), i.e., inlet setpoints of 150 °C and 300 °C and charging times of 1.5 h and 3 h. Vertical dashed lines mark the standby interval between charging and discharging.

Consistent with the analysis in Section 3.2, all panels in Figs. B1 and B2 show that (i) inner-radius sensors warm earlier and reach higher temperatures during charge, evidencing a core-to-wall gradient; (ii) the mid-height level tends to hold the highest temperatures due to lower end losses; (iii) longer charging time (3 h) promotes a more uniform distribution at the end of charge compared with 1.5 h; and (iv) during discharge, outer-radius sensors cool first while inner-radius sensors retain heat longer, revealing preservation of the previously established radial stratification. At the higher inlet setpoint (300 °C), early-time heating rates increase and standby losses become more noticeable, particularly near the periphery, in line with the larger wall-to-ambient temperature difference.

Reading guide. In both figures, legends identify the thermocouples associated with inner/outer radii and heights – Table 2 provides the mapping to physical positions. The inlet and outlet traces confirm that the heater control does not perturb the effective inlet at the bed entrance, thus the inner-radius measurements are representative of the imposed boundary condition. The appended profiles substantiate the core findings of Section 3.2 regarding core-to-wall gradients, end losses, and the evolution of radial stratification across operating conditions.

Data availability

Information may be made available on request. Additionally, datasets containing temperature profiles for runs A1 and R4 are available at https://doi.org/10.17632/fgn79fzg7x.1.

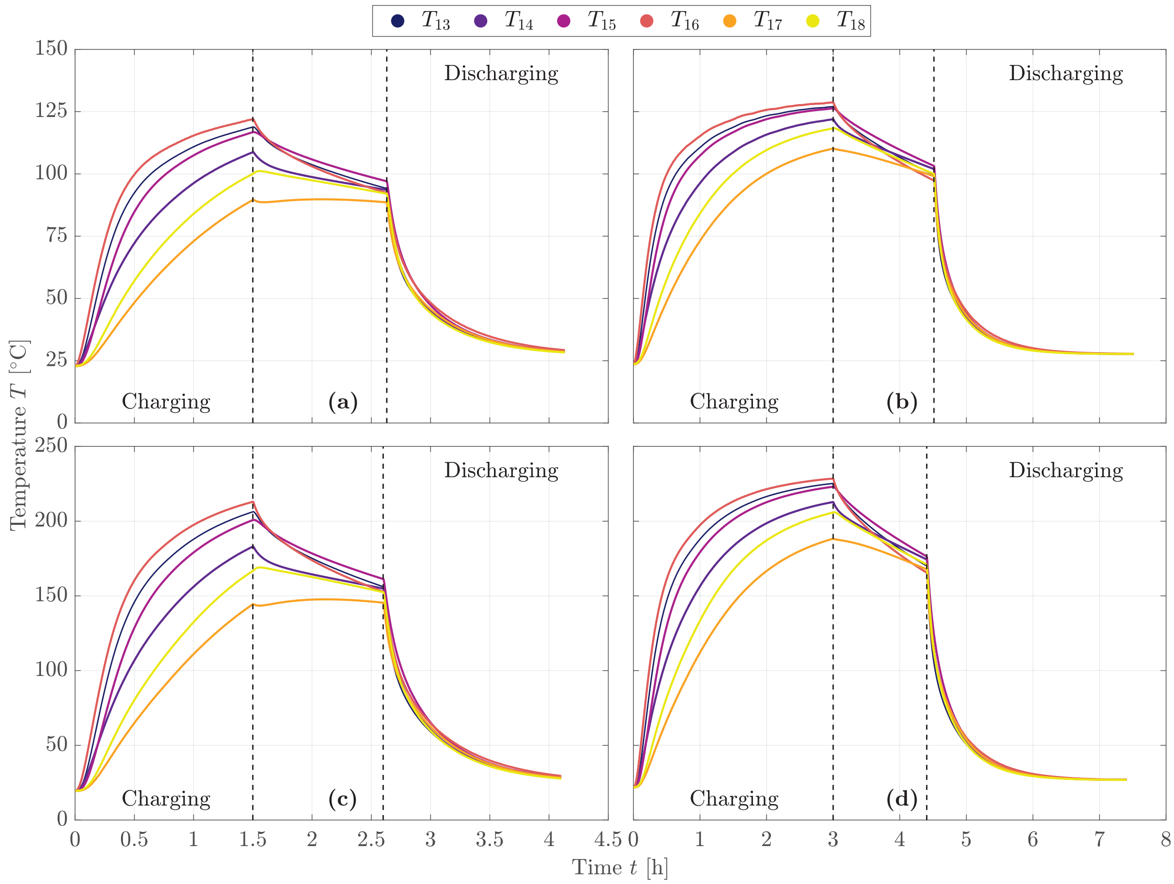


Fig. B2. Temperature profiles along the time for the R-PBTES device: (a) Run R1, (b) Run R2, (c) Run R3, and (d) Run R4.

References

[1] IEA, Key World Energy Statistics 2021, Tech. Rep., International Energy Agency, Paris, 2021.

[2] IRENA, World Energy Transitions Outlook 1.5° C Pathway, Tech. Rep., International Energy Agency, Abu Dhabi, UAE, 2021.

[3] S. Kalaiselvam, R. Parameshwaran, Chapter 3 - thermal energy storage technologies, in: S. Kalaiselvam, R. Parameshwaran (Eds.), Thermal Energy Storage Technologies for Sustainability, Academic Press, Boston, 2014, pp. 57–64, http://dx.doi.org/10.1016/B978-0-12-417291-3.00003-7.

[4] L. Miró, J. Gasia, L.F. Cabeza, Thermal energy storage (TES) for industrial waste heat (IWH) recovery: A review, Appl. Energy 179 (2016) 284–301, http://dx.doi.org/10.1016/j.apenergy.2016.06.147, URL https://linkinghub.elsevier.com/retrieve/pii/S0306261916309357.

[5] O. Abdur Rehman, V. Palomba, A. Frazzica, L.F. Cabeza, Enabling technologies for sector coupling: A review on the role of heat pumps and thermal energy storage, Energies 14 (24) (2021) 8195, http://dx.doi.org/10.3390/en14248195.

[6] A.M. Pantaleo, S. Trevisan, F. Matteucci, L.F. Cabeza, Innovation trends on high-temperature thermal energy storage to defossilize energy systems, J. Energy Storage 103 (2024) 114261, http://dx.doi.org/10.1016/j.est.2024.114261, URL https://linkinghub.elsevier.com/retrieve/pii/S2352152X24038477.

[7] H. Agalit, N. Zari, M. Maaroufi, Thermophysical and chemical characterization of induction furnace slags for high temperature thermal energy storage in solar tower plants, Sol. Energy Mater. Sol. Cells 172 (2017) 168–176, http://dx.doi.org/10.1016/j.solmat.2017.07.035.

[8] K. Knobloch, Y. Muhammad, M.S. Costa, F.M. Moscoso, C. Bahl, O. Alm, K. Engelbrecht, A partially underground rock bed thermal energy storage with a novel air flow configuration, Appl. Energy 315 (2022) 118931, http://dx.doi.org/10.1016/j.apenergy.2022.118931, URL https://linkinghub.elsevier.com/retrieve/pii/S0306261922003464.

[9] X. Zhuang, X. Xu, W. Liu, W. Xu, LCOE analysis of tower concentrating solar power plants using different Molten-Salts for thermal energy storage in China, Energies 12 (7) (2019) 1394.

[10] L.A. Weinstein, J. Loomis, B. Bhatia, D.M. Bierman, E.N. Wang, G. Chen, Concentrating solar power, Chem. Rev. 115 (23) (2015) 12797–12838, http://dx.doi.org/10.1021/acs.chemrev.5b00397.

[11] G. Zanganeh, A. Pedretti, S. Zavattoni, M. Barbato, A. Steinfeld, Packed-bed thermal storage for concentrated solar power – pilot-scale demonstration and industrial-scale design, Sol. Energy 86 (10) (2012) 3084–3098, http://dx.doi.org/10.1016/j.solener.2012.07.019.

[12] T. Esence, A. Bruch, S. Molina, B. Stutz, J.-F. Fourmigué, A review on experience feedback and numerical modeling of packed-bed thermal energy storage systems, Sol. Energy 153 (2017) 628–654, http://dx.doi.org/10.1016/j.solener.2017.03.032.

[13] I. González, C. Pérez-Segarra, O. Lehmkuhl, S. Torras, A. Oliva, Thermo-mechanical parametric analysis of packed-bed thermocline energy storage tanks, Appl. Therm. Energy 179 (2016) 1106–1122, http://dx.doi.org/10.1016/j.apenergy.2016.06.124.

[14] G. Zanganeh, A. Pedretti, A. Haselbacher, A. Steinfeld, Design of packed bed thermal energy storage systems for high-temperature industrial process heat, Appl. Energy 137 (2015) 812–822, http://dx.doi.org/10.1016/j.apenergy.2014.07.110.

[15] S. Trevisan, Y. Jemmal, R. Guedez, B. Laumert, Packed bed thermal energy storage: A novel design methodology including quasi-dynamic boundary conditions and techno-economic optimization, J. Energy Storage 36 (2021) 102441, http://dx.doi.org/10.1016/j.est.2021.102441, URL https://linkinghub.elsevier.com/retrieve/pii/S2352152X21001936.

[16] Brenmiller thermal energy storage, 2025, https://bren-energy.com/. (Accessed 22 September 2025).

[17] Polar night energy sand battery, 2025, https://polarnightenergy.com/. (Accessed 22 September 2025).

[18] The rondo heat battery, 2025, https://www.rondo.com/. (Accessed 22 September 2025).

[19] M. Hänchen, S. Brückner, A. Steinfeld, High-temperature thermal storage using a packed bed of rocks – Heat transfer analysis and experimental validation, Appl. Therm. Eng. 31 (10) (2011) 1798–1806, http://dx.doi.org/10.1016/j.applthermaleng.2010.10.034.

[20] M. Cascetta, G. Cau, P. Puddu, F. Serra, A study of a packed-bed thermal energy storage device: Test rig, experimental and numerical results, Energy Procedia 81 (2015) 987–994, http://dx.doi.org/10.1016/j.egypro.2015.12.157, 69th Conference of the Italian Thermal Engineering Association, ATI 2014.

[21] I. Ortega-Fernández, I. Loroño, A. Faik, I. Uriz, J. Rodríguez-Aseguinolaza, B. D'Aguanno, Parametric Analysis of a Packed Bed Thermal Energy Storage System,

Abu Dhabi, United Arab Emirates, 2017, p. 080021, http://dx.doi.org/10.1063/1.4984442.

[22] S.J. Erasmus, T.W. von Backström, M. Lubkoll, F. Dinter, Design and development of a next generation thermal rock bed storage experimental facility, AIP Conf. Proc. 2126 (1) (2019) 200017, http://dx.doi.org/10.1063/1.5117732.

[23] S. Trevisan, W. Wang, R. Guedez, B. Laumert, Experimental evaluation of an innovative radial-flow high-temperature packed bed thermal energy storage, Appl. Energy 311 (2022) 118672, http://dx.doi.org/10.1016/j.apenergy.2022.118672.

[24] F. Cabello Núñez, J. López Sanz, F. Zaversky, Analysis of steel making slag pebbles as filler material for thermocline tanks in a hybrid thermal energy storage system, Sol. Energy 188 (2019) 1221–1231, http://dx.doi.org/10.1016/j.solener.2019.07.036.

[25] I. Calderón-Vásquez, V. Segovia, J.M. Cardemil, R. Barraza, Assessing the use of copper slags as thermal energy storage material for packed-bed systems, Energy 227 (2021) http://dx.doi.org/10.1016/j.energy.2021.120370.

[26] J. McTigue, A. White, A comparison of radial-flow and axial-flow packed beds for thermal energy storage, Appl. Energy 227 (2018) 533–541, http://dx.doi.org/10.1016/j.apenergy.2017.08.179, Transformative Innovations for a Sustainable Future – Part III.

[27] M.E. Skuntz, R. Elander, M. Al Azawii, P. Bueno, R. Anderson, System efficiency of packed bed TES with radial flow vs. axial flow – influence of aspect ratio, J. Energy Storage 72 (2023) 108463, http://dx.doi.org/10.1016/j.est.2023.108463.

[28] H. Zhou, Z. Lai, K. Cen, Experimental study on energy storage performances of packed bed with different solid materials, Energy 246 (2022) 123416, http://dx.doi.org/10.1016/j.energy.2022.123416.

[29] M. Prenzel, V. Danov, S. Will, L. Zigan, T. Barmeier, J. Schäfer, Thermo-fluid dynamic model for horizontal packed bed thermal energy storages, Energy Procedia 135 (2017) 51–61, http://dx.doi.org/10.1016/j.egypro.2017.09.486, 11th International Renewable Energy Storage Conference, IRES 2017, 14-16 March 2017, Düsseldorf, Germany.

[30] M.M. Al-Azawii, D. Jacobsen, P. Bueno, R. Anderson, Experimental study of thermal behavior during charging in a thermal energy storage packed bed using radial pipe injection, Appl. Therm. Eng. 180 (2020) 115804, http://dx.doi.org/10.1016/j.applthermaleng.2020.115804.

[31] S. Soprani, F. Marongiu, L. Christensen, O. Alm, K.D. Petersen, T. Ulrich, K. Engelbrecht, Design and testing of a horizontal rock bed for high temperature thermal energy storage, Appl. Energy 251 (2019) 113345, http://dx.doi.org/10.1016/j.apenergy.2019.113345.

[32] M.v.d. Heyde, Electric Thermal Energy Storage Based on Packed Beds for Renewable Energy Integration (Ph.D. thesis), Technische Universität Hamburg, 2022.

[33] W.H. Gerstle, N.R. Schroeder, L.P. McLaughlin, C.K. Ho, H.F. Laubscher, S. Kao, Experimental testing and computational modeling of a radial packed bed for thermal energy storage, Sol. Energy 264 (2023) 111993, http://dx.doi.org/10.1016/j.solener.2023.111993.

[34] J.F. Tuttle, N. White, K. Mohammadi, K. Powell, A novel dynamic simulation methodology for high temperature packed-bed thermal energy storage with experimental validation, Sustain. Energy Technol. Assess. 42 (2020) 100888, http://dx.doi.org/10.1016/j.seta.2020.100888.

[35] M.M. Al-Azawii, S.F. Alhamdi, S. Braun, J.-F. Hoffmann, N. Calvet, R. Anderson, Experimental study on packed-bed thermal energy storage using recycled ceramic as filler materials, J. Energy Storage 44 (2021) 103375, http://dx.doi.org/10.1016/j.est.2021.103375.

[36] Z. Lai, K. Cen, H. Zhou, Applicability of coal slag for application as packed bed thermal energy storage materials, Sol. Energy 236 (2022) 733–742, http://dx.doi.org/10.1016/j.solener.2022.03.031.

[37] J. Ochmann, Ł. Bartela, K. Rusin, M. Jurczyk, B. Stanek, S. Rulik, S. Waniczek, Experimental studies of packed-bed thermal energy storage system performance, J. Power Technol. 102 (1) (2022) 37–44.

[38] W. Wang, X. He, Y. Shuai, J. Qiu, Y. Hou, Q. Pan, Experimental study on thermal performance of a novel medium-high temperature packed-bed latent heat storage system containing binary nitrate, Appl. Energy 309 (2022) 118433, http://dx.doi.org/10.1016/j.apenergy.2021.118433.

[39] Y. Wu, H. Geng, G. Hao, D. Li, Experimental study on heat exchange efficiency of rock bed heat storage system based on broken rock mass, 2022, http://dx.doi.org/10.2139/ssrn.4104998, SSRN.

[40] E. Alonso, E. Rojas, R. Bayón, Experimental and numerical study of an air solid thermocline thermal energy storage system operating at high temperature, AIP Conf. Proc. 2445 (1) (2022) 160001, http://dx.doi.org/10.1063/5.0087655.

[41] J. Hoffmann, T. Fasquelle, V. Goetz, X. Py, A thermocline thermal energy storage system with filler materials for concentrated solar power plants: Experimental data and numerical model sensitivity to different experimental tank scales, Appl. Therm. Eng. 100 (2016) 753–761, http://dx.doi.org/10.1016/j.applthermaleng.2016.01.110.

[42] A. Bruch, S. Molina, T. Esence, J. Fourmigué, R. Couturier, Experimental investigation of cycling behaviour of pilot-scale thermal oil packed-bed thermal storage system, Renew. Energy 103 (2017) 277–285, http://dx.doi.org/10.1016/j.renene.2016.11.029.

[43] X. Yang, X. Yang, F.G. Qin, R. Jiang, Experimental investigation of a molten salt thermocline storage tank, Int. J. Sustain. Energy 35 (6) (2016) 606–614, http://dx.doi.org/10.1080/14786451.2014.930465.

[44] S. Rodat, A. Bruch, N. Dupassieux, N.E. Mourchid, Unique fresnel demonstrator including ORC and thermocline direct thermal storage: Operating experience, Energy Procedia 69 (2015) 1667–1675, http://dx.doi.org/10.1016/j.egypro.2015.03.127, International Conference on Concentrating Solar Power and Chemical Energy Systems, SolarPACES 2014.

[45] E. Alami, Asbik, Agalit, Identification of natural rocks as storage materials in thermal energy storage (TES) system of concentrated solar power (CSP) plants– A review, Sol. Energy Mater. Sol. Cells 217 (2020) 110599, http://dx.doi.org/10.1016/j.solmat.2020.110599.

[46] D. Fernandes, F. Pitié, G. Cáceres, J. Baeyens, Thermal energy storage: "how previous findings determine current research priorities", Energy 39 (1) (2012) 246–257, http://dx.doi.org/10.1016/j.energy.2012.01.024, Sustainable Energy and Environmental Protection 2010.

[47] M.E. Navarro, M. Martínez, A. Gil, A.I. Fernández, L.F. Cabeza, R. Olives, X. Py, Selection and characterization of recycled materials for sensible thermal energy storage, Sol. Energy Mater. Sol. Cells 107 (2012) 131–135, http://dx.doi.org/10.1016/j.solmat.2012.07.032.

[48] Gil, Calvet, Ortega, Risueño, Faik, et al., Characterization of a by-product from steel industry applied to thermal energy storage in concentrated solar power, Proc. Est. Acad. Sci. Biol. Ecol. (2014).

[49] Y. Rong, S.Y. Huang, H. Zhou, Experimental study on storage performance of packed bed solar thermal energy storage system using steel slag, J. Energy Storage 78 (2024) 110042, http://dx.doi.org/10.1016/j.est.2023.110042.

[50] P.A. Curto, G. Stern, Thermal storage using slag, Altern. Energy Sources;(United States) 1 (CONF-801210-) (1980).

[51] G. Alkan, P. Mechnich, H. Lucas, N. Knoblauch, M. Sommerfeld, F. Flucht, J. Pernpeintner, D. Sergeev, M. Müller, B. Friedrich, Assessment of metallurgical slags as solar heat absorber particles, Minerals 12 (2) (2022) 121, http://dx.doi.org/10.3390/min12020121.

[52] I. Ortega-Fernández, N. Calvet, A. Gil, J. Rodríguez-Aseguinolaza, A. Faik, B. D'Aguanno, Thermophysical characterization of a by-product from the steel industry to be used as a sustainable and low-cost thermal energy storage material, Energy 89 (2015) 601–609, http://dx.doi.org/10.1016/j.energy.2015.05.153.

[53] Y. Wu, J. Li, M. Wang, H. Wang, Y. Zhong, Y. Zhao, M. Wei, Y. Li, Solar salt doped by MWCNTs as a promising high thermal conductivity material for CSP, RSC Adv. 8 (2018) 19251–19260, http://dx.doi.org/10.1039/C8RA03019G.

[54] R. Jacob, D. Sergeev, M. Müller, Valorisation of waste materials for high temperature thermal storage: a review, J. Energy Storage 47 (2022) 103645, http://dx.doi.org/10.1016/j.est.2021.103645.

[55] DOWTHERM™ A heat transfer fluid, 2001, https://www.dow.com/en-us/pdp.dowtherm-a-heat-transfer-fluid.238000z.html?productCatalogFlag=1. (Accessed 4 January 2023).

[56] C.W. Robak, T.L. Bergman, A. Faghri, Economic evaluation of latent heat thermal energy storage using embedded thermosyphons for concentrating solar power applications, Solar Energy 85 (10) (2011) 2461–2473, http://dx.doi.org/10.1016/j.solener.2011.07.006.

[57] I. Calderón-Vásquez, I. Wolde, V. Segovia, F. Battisti, J.M. Cardemil, R. Escobar, Modelling the temperature distribution in a horizontal packed-bed thermal energy storage system with copper slag as filler material, J. Energy Storage 125 (2025) 116769, http://dx.doi.org/10.1016/j.est.2025.116769, URL https://www.sciencedirect.com/science/article/pii/S2352152X25014823.

[58] I. Calderón-Vásquez, F.G. Battisti, R. Escobar, J.M. Cardemil, Operational dynamics of packed-bed thermal energy storage: A novel approach to monitor its thermal state, J. Energy Storage 129 (2025) 116954, http://dx.doi.org/10.1016/j.est.2025.116954, URL https://www.sciencedirect.com/science/article/pii/S2352152X25016676.

[59] V. Segovia, I. Calderón-Vásquez, J.M. Cardemil, Statistical modeling of copper slag heat capacity for thermal storage, Sustain. Energy Technol. Assess. 82 (2025) 104458, http://dx.doi.org/10.1016/j.seta.2025.104458, URL https://www.sciencedirect.com/science/article/pii/S2213138825002899.

[60] I. Wolde, I. Calderón-Vásquez, M. Molina, N. Pailahueque, J.M. Cardemil, Parametric analysis of a modular solar drying and packed bed thermal energy storage system, Sol. Energy 300 (2025) 113764, http://dx.doi.org/10.1016/j.solener.2025.113764, URL https://www.sciencedirect.com/science/article/pii/S0038092X25005274.

[61] M.A. David-Hernández, I. Calderon-Vásquez, F.G. Battisti, J.M. Cardemil, A. Cazorla-Marín, Design and assessment of a concentrating solar thermal system for industrial process heat with a copper slag packed-bed thermal energy storage, Appl. Energy 376 (2024) 124280, http://dx.doi.org/10.1016/j.apenergy.2024.124280, URL https://www.sciencedirect.com/science/article/pii/S0306261924016635.

[62] F. Battisti, L. de Araujo Passos, A. da Silva, Performance mapping of packed-bed thermal energy storage systems for concentrating solar-powered plants using supercritical carbon dioxide, Appl. Therm. Eng. 183 (2021) 116032, http://dx.doi.org/10.1016/j.applthermaleng.2020.116032, URL https://www.sciencedirect.com/science/article/pii/S1359431120335134.